

BATTERY STORAGE AND NATURAL GAS GENERATOR MARKET POWER

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Abstract

Utility-scale batteries are often valued for their ability to integrate renewables. They also compete directly with the fossil generators that exercise market power when demand peaks, but the value of this competition has not been quantified. I study this competitive effect in Texas, where more than 7 GW of batteries entered after 2021. Using unit-level data on generator output, offers, costs, and local battery entry, I show that batteries make the residual demand of nearby gas generators more elastic, displace their evening output, and lower their evening markups. An additional 100 MWh of local storage reduces gas operating profits by 6 percent, and between 71 percent and all of this loss is market power rent. Gas generators keep setting prices, but at smaller margins. The resulting \$47 million annual transfer flows mainly to consumers, while shifting energy across hours creates little value at marginal cost. Batteries earn their arbitrage revenue from the markups they erode, so competition limits the private return to storage.

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1 Introduction

Battery storage is viewed as an important technology for the energy transition because it complements intermittent renewable generation. By shifting energy across time, storage smooths the variability of wind and solar output and allows renewables to supply a larger share of generation (Karaduman, 2023; Andres-Cerezo and Fabra, 2025). The welfare gains from this complementarity are typically framed as reductions in environmental externalities when storage-backed renewables displace fossil-fuel generation. An underappreciated role of storage, however, is that it competes directly with the conventional generators. Fossil-fuel plants, specifically the marginal units during peak demand hours, earn markups that reflect their market power (Borenstein, Bushnell and Wolak, 2002; Woerman, 2023). Because battery dispatch also targets the same peak hours, their entry makes the residual demand facing the incumbents more elastic and mitigates their market power. This competitive effect is distinct from the reduction in external costs, and it has not been quantified. Whether it matters in aggregate, and who captures the surplus it creates, are important empirical questions.

In this paper, I quantify how battery storage entry affects the market power of incumbent natural gas generators. I study the Texas electricity market (ERCOT), an energy-only market in which frequent transmission congestion allows local generators to extract scarcity rents. ERCOT provides an ideal setting for this analysis. More than 7 GW of battery capacity has been deployed in ERCOT since 2021, a scale of market penetration substantially larger than those evaluated in prior studies.¹ To identify the competitive effects of this entry, I construct a unit-level panel that links each gas generator’s output, offers, and marginal cost to storage entry in their local market. To account for congestion, I define local markets using price divergence and high-resolution grid topology (Arkolakis and Walsh, 2024; Mercadal, 2022; Woerman, 2023), and measure the slope of each generator’s residual demand from the offers of its local rivals.

To guide the empirical analysis, I develop a simple two-period model where a strategic natural gas generator competes with a price-taking storage fleet. Storage affects the generator’s markup through two channels. Through the *level channel*, storage shifts the generator’s residual demand, lowering it when batteries discharge and raising it when they charge. Through the *slope channel*, storage makes residual demand flatter, which raises the cost of withholding output. The model’s prediction maps each channel to an estimable parameter. The generator’s optimal output is linear in its marginal cost, with an intercept set by the level of residual demand and a slope set by its elasticity. Storage therefore enters output through two terms: a direct effect that captures displacement, and an interaction with cost that captures the slope channel, because a flatter residual demand displaces higher-cost generators more. The markup rule then shows that storage lowers markups through both channels in discharge hours, while its effect in charge hours is ambiguous.

My empirical strategy follows from the model prediction. Because the level channel reverses sign between charge and discharge hours, I estimate the marginal effect for each hour of the day rather

¹This capacity is roughly ten to one hundred times the storage analyzed in Butters, Dorsey and Gowrisankaran (2025), Karaduman (2023), and Kirkpatrick (2025).

than pooling across the day. Because the first-order condition holds only when a generator runs between its operating limits, I estimate the output rule on those hours and instrument marginal cost with the product of the pre-entry heat rate and the natural gas price. Because the cost interaction identifies the slope channel only in sign, I also measure the slope of each generator’s residual demand directly from the offers of its local rivals. Finally, because both channels lower operating profit, a fall in profit alone does not show that storage reduces market power. I further decompose profit into market power rent and competitive rent and compare it to benchmark competitive price.

I estimate these model parameters leveraging the staggered entry of storage capacity across ERCOT. As storage enters, incumbent natural gas generators are exposed to different levels of new competition across counties. My research design compares generators facing local battery entry with generators in counties that storage has not yet reached, before and after entry, and it uses the growing installed capacity in each county as the intensity of exposure. I carry the model’s output rule into this design by interacting local storage with each generator’s predetermined cost, so the estimates recover the same displacement and cost gradient as the model. Exploiting this spatial and temporal variation, I find four main results.

First, storage entry erodes the market power of incumbent natural gas generators during the evening peak hours. An additional 100 MWh of local storage reduces the markup of local gas generators by \$2.69 per MWh at 7 PM and by \$1.49 per MWh on average over the evening peak, about 5 percent of the pre-entry evening markup. Markups do not change overnight or in the morning, when batteries charge. Operating profit falls by \$5.33 per MW-day, about 6 percent of the pre-entry average. The reduction in market power grows with the installed dose, and the event study estimates provide no evidence of pre-entry trends.

Second, using the computed residual demand slope from the bid data, my estimates imply that residual demand of nearby gas generators becomes more elastic after storage enters. The effect is most pronounced for combined cycle generators in the evening. Storage also displaces gas output in the afternoon and evening, and it displaces higher-cost generators more, as the model predicts. My instrumental variables estimate of the generator’s optimal output rule confirms that output responds to cost as the model requires. These results hold when I control for the solar and wind capacity that often enters alongside storage.

Third, the lost profit is mainly market power rent rather than a lower competitive price. I compare the realized price with the competitive price that a cost-based dispatch of the same gas output would produce within the local market, following the approach of [Borenstein, Bushnell and Wolak \(2002\)](#). I use a more conservative lower bound of competitive price by using the cost of the generator that ERCOT identifies as the price-setter. Market power rent accounts for between 71 percent and all of the profit loss under the two benchmarks. Gas generators also continue to set the price as often as before storage entry. If storage lowered gas profits only by adding supply, it would push cheaper units or storage itself to the margin. Instead, gas generators keep setting the price at a smaller margin above cost, so storage disciplines their pricing rather than only adding supply.

Finally, I quantify the welfare implications of these competitive effects. The competitive effect of storage is mainly a transfer. Local storage lowered gas operating profits by about \$47 million per year, of which \$33 million to \$54 million was market power rent. Consumers gained about \$50 million from lower prices, although this estimate is imprecise. The efficiency gain from shifting energy is small. Valued at marginal cost, the energy arbitrage across hours is negligible, because the cost spread between charging and discharging hours barely covers round-trip losses. Valued at market prices, energy arbitrage earns storage owners \$5.8 million per year. The difference is the markup of gas generators, so storage earns its arbitrage revenue from the markups it erodes, and competition limits its private return from arbitrage. Energy arbitrage is also a small part of what storage earns. Storage owners earned another \$35.6 million per year from ancillary services, 86 percent of their total revenue of \$41.4 million, or \$55 per kW-year.² Even this total covers only about a quarter of the annualized capital cost of storage, and the measured social benefits, including avoided emissions at their upper bound, also fall short of it. At current penetration, the case for storage therefore rests on its competitive effect and its reliability services rather than on energy arbitrage.

This paper contributes to three strands of literature. First, it contributes to the broad industrial organization literature on the effect of entry on incumbent firm behavior. A body of empirical work shows that incumbents respond to the threat or realization of entry across industries, from retail supercenters (Arcidiacono et al., 2020; Busso and Galiani, 2019) and retail gasoline (Taylor and Muehlegger, 2026) to airlines (Goolsbee and Syverson, 2008) and pharmaceuticals (Ellison and Ellison, 2011). The papers closest to my setting are Arcidiacono et al. (2020) and Taylor and Muehlegger (2026), who use high-frequency microdata to evaluate how incumbents adjust to entry. I extend this framework to wholesale electricity markets, where I observe the incumbent’s output, residual demand, and realized margin at a more granular level. Moreover, the entrant in my setting is not a symmetric rival because storage buys when incumbents face little competition and sells when they face the most, so its competitive effect varies across hours of the day, and its own profit depends on the markups it erodes.

Second, this paper contributes to the literature on competition and market power in wholesale electricity markets. A large body of work documents how generators exercise market power under transmission constraints and localized congestion (Borenstein, Bushnell and Wolak, 2002; Hortaçsu and Puller, 2008; Ito and Reguant, 2016; Woerman, 2023), and recent studies show that transmission expansion (Ryan, 2021; Doshi, 2024), financial arbitrageurs (Mercadal, 2022), and the price exposure of renewables (Fabra and Imelda, 2023) can reduce it. Closer to this paper, two recent studies examine how large-scale solar entry affects conventional generators. Jha and Leslie (2025) show that it softens evening competition through start-up costs, while Bushnell and Novan (2021) show that it shifts the marginal technology toward less efficient units. I study a different large-scale

²These figures are in line with industry estimates. Modo Energy reports that the average ERCOT battery earned \$141 per kW in 2022, \$192 per kW in 2023, and \$55 per kW in 2024, and that energy arbitrage provided about 15 percent of battery revenue in 2023 (Vermillion, 2024, 2025). My average spans 2018 to 2024, excludes Winter Storms Uri and Elliott, and is in 2020 dollars, which explains its lower level.

entry, storage. I show that storage disciplines market power through a similar residual demand channel that this literature identifies as central to transmission-driven discipline.

Finally, this paper speaks to the growing literature on the economics of utility-scale storage. This work has largely quantified the aggregate, market-level impact of storage, documenting how it substitutes for or complements renewable capacity (Karaduman, 2023; Andres-Cerezo and Fabra, 2025), lowers intraday price spreads (Lamp and Samano, 2022), shifts long-run investment equilibria (Butters, Dorsey and Gowrisankaran, 2025), and relieves local congestion (Kirkpatrick, 2025). Less is known about how entry changes the behavior of incumbents. Closest to this paper, Reaser and Taylor (2025) uses the entry of storage in ERCOT to estimate its effect on locational marginal prices, and builds a structural dispatch model that computes counterfactual prices without storage offers. I leverage the same source of variation focusing on the marginal generators and measure how storage changes the output, residual demand, and markups. I further separate the market power component of their lost profit from the effect of a lower competitive price. Other recent papers study different margins of storage entry. Dudek (2026) studies how storage reduces renewable curtailment in California, and Visaggio (2025) studies how ownership structure shapes the market incentives of batteries paired with renewables in Texas. I complement these papers by focusing on its competition with gas generators. To my knowledge, this paper is the first to use unit-level storage dispatch to trace the channel from an individual storage unit’s entry to the behavior of incumbent gas generators. While prior work characterizes the theoretical price impact of storage (Andrés-Cerezo and Fabra, 2023), I provide causal evidence that observed storage entry erodes realized markups. I also use a welfare framework that separates transfers from net social gains, and show that the private return to storage arbitrage depends on the market power it disciplines.

The remainder of this paper is organized as follows. Section 2 provides an overview of the Texas electricity market. Section 3 provides the data and descriptive statistics on natural gas and storage units. Section 4 develops the theoretical model. Section 5 presents the empirical strategy. Section 6 presents the effects on output, residual demand, markups, and operating profit. Section 7 quantifies the welfare and distributional effects, and Section 8 concludes.

2 Storage and The Texas Electricity Market

In this section, I provide an overview of the Texas electricity market, highlighting the unique features of ERCOT that make it an ideal setting to study the impact of storage entry on natural gas generator market power.

The Texas electricity market provides an ideal setting to study the competitive interactions between storage and natural gas generators. The Electric Reliability Council of Texas (ERCOT) manages this grid, delivering power to more than 26 million customers, about 90% of the state’s electric load, and operating a wholesale market that schedules the dispatch of over 1,500 generators in real time. Because it has limited interconnection with other regions, the market is relatively insulated from external shocks, creating a highly contained environment for empirical analysis.

This system is particularly useful to study because of its generation mix; natural gas accounts for a dominant 44.3% of the generation capacity, followed by wind (25.2%) and solar (13.2%) (ERCOT, 2025).³ At the same time, massive recent growth in battery storage capacity is actively altering these established competition dynamics. Furthermore, because ERCOT operates as an energy-only market with no capacity payments, it creates strong financial incentives for generators to respond strategically to real-time conditions.

Storage Participation and Competition. The massive growth of utility-scale battery storage in ERCOT introduces new competitors to incumbent natural gas generators. Storage units operate as fully dispatchable assets that offer operational flexibility, functioning as generators when exporting and as load when charging. They participate in both the day-ahead and real-time markets providing both energy and ancillary services.⁴ Figure 1 documents this expansion, with ERCOT storage capacity growing to over 7.6 GW since 2021.⁵

Because battery storage can respond quickly to price signals, these units can precisely target their dispatch during high-revenue periods where incumbent generators exercised market power. Panel B documents this pattern where storage discharges predominantly during the evening peak (hours 16–20). Panels C and D provide further descriptive evidence of competition during these hours: the share of storage units clearing at peak prices has increased, while the corresponding share of marginal natural gas generators has declined as more storage capacity enters. Before formally estimating the magnitude of this competitive effect, I first outline the features of the ERCOT market design that facilitate these interactions.

Market Design and Market Power. ERCOT operates an energy-only market that runs sequentially: the day-ahead market and the real-time market. Unlike PJM, MISO, and ISO-NE, ERCOT has no centralized capacity market, and generators are compensated based on the electricity and ancillary services they provide.⁶ Price is determined at each node in the grid, with the locational marginal price (LMP) at each settlement point reflecting generation costs, transmission constraints, and line losses.⁷ Generators first submit offers for each hour of the next day in the day-ahead market. The real-time market then runs every five minutes for generators to adjust

³Figure E.19 shows the spatial distribution of each unit based on their fuel type in ERCOT. Table D.7 summarizes the generator units in ERCOT territory as of the end of 2024, while Figure E.21 details the installed capacity over my sample period: Natural gas units dominate the installed capacity (38.9% or 65.23 GW), followed by wind and solar PV units, respectively. Natural gas and coal capacity remain roughly the same over my sample, while solar PV and wind capacity grow significantly by about 25 GW and 20 GW.

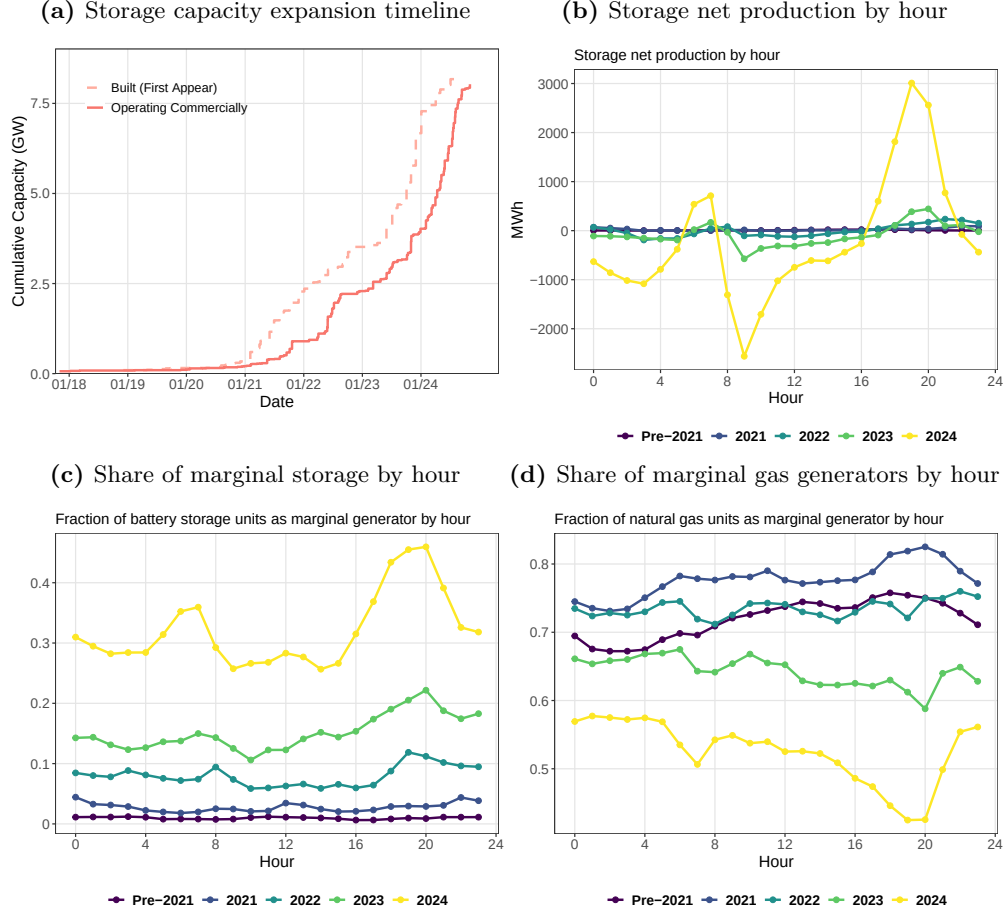
⁴Because they serve as both generation and load, batteries can provide a wide range of regulation and reserve services. During summer 2023, for example, ERCOT Contingency Reserve Service (ECRS) accounted for more than 75% of total battery revenues, whereas real-time energy settlement contributed just under 15% (Miller and Nakajima, 2024). Figure E.20 illustrates how a single storage resource can simultaneously fulfill multiple AS responsibilities.

⁵In Q2 2025, ERCOT’s installed storage capacity surpassed California’s. A fundamental difference between the two markets is that ERCOT’s expansion is driven by price signals and efficient interconnection queues, rather than the state-level procurement mandates or subsidies seen in California (Mathrani, 2025; Ahmed et al., 2026).

⁶Ancillary services in ERCOT include regulation (REGUP, REGDN) and operating reserves (Responsive Reserve [RRS], Non-Spin [NSPIN], and ERCOT Contingency Reserve Service [ECRS]).

⁷Each generator is assigned to a settlement point and receives the LMP at that location.

Figure 1: Storage expansion and its competition trends with natural gas generators in ERCOT



Notes: Panel A shows the cumulative storage capacity built and commercially operating over time. Panel B shows the aggregate storage net production pattern by hour. Net production is defined as aggregate storage discharge less aggregate storage charge. Panel C shows the share of storage units that receive the highest offered price in each hour. Panel D shows the share of natural gas generators that receive the highest offered price in each hour. Section B.1 provides the same yearly trend in hourly demand, net demand, aggregate solar, wind, and natural gas generation.

their output from their day-ahead schedule to account for changes in realization of demand and renewable output. The Security Constrained Economic Dispatch (SCED) then clears the real-time market and sets the real-time LMPs.

Transmission congestion frequently splits the ERCOT grid into isolated local markets, allowing even small generators to become pivotal and exercise market power through economic withholding (LaRiviere and Lyu, 2022; Fell, Kaffine and Novan, 2021; Woerman, 2023; Doshi, 2024). Because congestion isolates firms in these local markets from broader sets of competitors on the other side of constraints, firms can exercise market power by raising offer prices above marginal cost causing the market to clear at higher LMPs. Consequently, the deployment of battery storage presents a competitive threat to these specific withholding strategies within that local market.

This incentive to withhold is driven by the energy-only structure and enabled by permissive regulatory price caps. Without centralized capacity payments, generators face substantial revenue

uncertainty during normal operating hours and must rely on securing outsized scarcity rents during tight, congested intervals to cover their fixed operating costs. ERCOT provides firms with immense latitude to capture these scarcity rents through flexible bidding rules and historically loose price caps of up to \$9,000/MWh, lowered to \$5,000/MWh in 2022.⁸ Unlike other centralized electricity markets that tightly regulate offers and impose rigid caps around \$1,000/MWh or less to mitigate market power (Joskow, 2022), ERCOT’s regulations explicitly permit generators to raise prices sharply when local scarcity conditions emerge.

3 Data and Descriptive Evidence

This section describes the data sources and the construction of the panel dataset used to analyze the impact of storage entry on natural gas generator market power. I first outline the sample construction and the empirical definitions of local markets, marginal costs, and markups. I then provide descriptive statistics for natural gas generators and storage units.

3.1 Data Sources and Sample Construction

I combine multiple datasets to construct a panel at the generator-by-hour level covering January 1, 2018 to December 1, 2024. The main components are the generator and storage unit characteristics, supply offers and bids, the market outcomes, and marginal costs. The novel component of this dataset is that I observe each storage unit’s power capacity, storage duration, bid/offer curves, and unit-level charge and discharge quantity.

Generator and battery storage data. I identify unique generators from ERCOT’s 60-day Security Constrained Economic Dispatch (SCED) Report and the Monthly Outlook for Resource Adequacy (MORA) Report.⁹ For all generators, I define effective capacity as the maximum production observed in the SCED Report, and I use the first date a generator produces electricity outside of testing as its initial commercial operating date (COD).

Because ERCOT reports power capacity but not energy capacity for battery storage, I complement this data using EIA Form 860M, the MODO Energy Battery Buildout Report, and other public sources to determine each unit’s storage duration. Furthermore, because storage acts as both a generator and a load, it operates under different names when submitting supply offers versus load bids; I manually link these identifiers for each storage unit. Finally, I use EIA Form 860 to assign the load zone and county for each generator, and the Decision Making Entity from the SCED MORA Report to identify the parent company.¹⁰

⁸Under FERC Order 831, non-ERCOT markets allow offer caps to increase to \$2,000/MWh to set the market clearing price, but strictly on the condition that these higher offers are justified by verified short-run marginal costs.

⁹Two or more generators may be associated with a single storage or generation unit. A generator’s name might also change before and after testing. For combined-cycle units, I use the maximum production from the generator combination with the highest possible output as the effective capacity.

¹⁰I manually crosscheck this ownership data with EIA Form 860 and employ AI tools to determine the parent company for each generator. Generators owned by multiple companies are coded under a single parent company.

Market outcome data. I collect 15-minute interval data on offer and bid curves, dispatch instructions, operating status, and the identity of marginal generators directly from the SCED reports. I also extract 5-minute settlement point LMPs, mapping them to individual generators and aggregating them to 15-minute intervals. I then aggregate these market outcomes to the hourly level, constructing variables for total hourly generation, average LMP, and an indicator for whether a generator was marginal at any point during the hour.

To define the local markets created by transmission congestion, I group generators based on LMP divergence and network connectivity at each interval, following [Mercadal \(2022\)](#) and [Woerman \(2023\)](#). I first group generators whose LMPs are within \$1/MWh or 1% of the larger price. I then verify transmission connectivity within these groups using transmission lines and substation shapefiles collected by [Arkolakis and Walsh \(2024\)](#). Generators sharing both similar LMPs and grid connections constitute a local market. Figure B.3 shows examples of market definition for two intervals, one with no congestion and one with congestion. The number of local markets varies from 1 to 137 across intervals, with a median of 8.¹¹ Within these local markets, I estimate residual demand for each generator by stacking the bids of all other generators, computing the slope at the average LMP using an Epanechnikov kernel for bids within \$5/MWh of the LMP. The resulting slope represents the density of rival supply, measured in MW per (\$/MWh). This captures how many additional MW of rival capacity would enter the local market if the LMP were to increase by one dollar.

Marginal cost. I construct the marginal cost of natural gas generators as the sum of variable fuel cost plus emission permit costs following [Hausman \(2025\)](#) and [Doshi \(2024\)](#):

$$c_{it} = \underbrace{hr_{it} \cdot p_{it}^{Fuel}}_{\text{Fuel Cost}} + \underbrace{ER_{it}^{SO_2} p_t^{SO_2} + ER_{it}^{NO_x} p_t^{NO_x}}_{\text{Emission Cost}}.$$

I use hourly generation data from SCED and collect information on heat content and emissions to compute heat rates and emission rates at the plant-by-type-by-week level from the Environmental Protection Agency’s (EPA) Continuous Emission Monitoring System (CEMS) from 2018 to 2024.¹² To compute fuel cost, I collect daily natural gas price data from S&P Capital IQ. Natural gas units source their fuel from different hubs. I use the natural gas price location information for each unit from SNL Energy.¹³ To compute emission cost, I collect permit prices for SO₂ and NO_x from S&P

¹¹Figure B.2 presents the distribution within my sample and across hours by year of sample. The number of local markets has increased over the years as more generators are added to the ERCOT grid.

¹²Heat rate and emission rate vary depending on the operating condition of a generator, as discussed by [Demirer and Karaduman \(2026\)](#). I compute the heat rate by dividing the weekly total heat content and emissions for that week by the weekly total net generation for each plant-type combination. For units with no observation in CEMS, I assign the average heat rate for the plant with the same Type (Combined Cycle, Simple Cycle, or Gas Steam Turbine). I winsorize the extreme values to be within the reported heat rate range for the same generator type from EPA eGrid, to account for unreasonably high and low values from CEMS.

¹³For units missing this information in the S&P Capital IQ database, I assign Waha Hub for units in the West load zone, NGPL TX-OK for units in the North load zone, TETCO STX for units in the South load zone, and Katy for units in the Houston load zone.

Capital IQ and apply the emission cost to units subject to the Acid Rain SO₂ Program and CSAPR NO_x Program.

Markup. I use realized markup as the primary outcome variable to measure market power, defined as:

$$\mu_{it} = \mathbf{1}[q_{it} > 0] (p_{it} - c_{it}).$$

This markup is zero when the generator is not producing electricity, and equal to the difference between its LMP p_{it} and marginal cost c_{it} when it is producing. This construction captures both the extensive margin (whether the generator is producing) and the intensive margin (how much markup it earns when producing). This is consistent with the electricity market literature that measures markup over all unit-time rather than only when a unit is dispatched (Bushnell, Mansur and Saravia, 2008; Borenstein, Bushnell and Wolak, 2002; Reguant, 2014).

Other data. To control for local weather conditions, I construct a balanced hourly panel of local temperature using station data from the NOAA Integrated Surface Database (ISD) (NOAA, 2024), interpolating minor gaps and assigning local temperatures to each power plant based on its nearest weather station. I also construct hourly renewable energy potential at the county level from the global horizontal irradiance data from National Solar Radiation Database (NSRDB) NREL (2024) and wind speed from ERA5 global reanalysis data (C3S, 2018).

Sample construction. I limit my estimation sample to marginal natural gas generators, defined at the generator-by-hour level. To prevent endogenous selection from storage shifting the merit order, I restrict the sample to generator-hour pairs that set the locational marginal price prior to local storage entry, requiring at least 100 pre-treatment observations to ensure I capture established operational patterns. I exclude the period of Winter Storms Uri and Elliott from my estimation sample. Next, I describe the gas generator’s characteristics.

3.2 Descriptive Statistics on Natural Gas Generators

Table 1 presents the descriptive statistics on natural gas generators. My sample consists of 558 natural gas generators (211 Simple Cycle, 298 Combine Cycle, and 49 Steam Turbine generators), with an average capacity of 366 MW and an average marginal cost of \$31/MWh. Geographically, these generators cluster heavily in the Houston, South, and North load zones.

Columns (1)–(3) detail within day variation in market outcomes: later hours (12 PM–12 AM) which include the evening peak hours consistently feature higher LMPs, capacity factors, and markups compared to early hours. Figure B.4 presents a detailed daily pattern of these outcomes over the years. These elevated afternoon and evening markups link directly to tighter market conditions, specifically, smaller local market sizes and steeper residual demand that enhance generator market power. Columns (4)–(6) present identical statistics for generators that share the same

Table 1: Summary statistics of natural gas generators

	All			Treated			Never-Treated		
	All	12AM –12PM	12PM –12AM	All	12AM –12PM	12PM –12AM	All	12AM –12PM	12PM –12AM
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Panel A: Market Outcome and Characteristics (2018–2021)									
LMP (\$/MWh)	28.7 (79.2)	21.7 (38.4)	35.7 (104.7)	28.9 (80.7)	21.9 (39.7)	36.0 (106.5)	27.8 (73.2)	21.1 (32.9)	34.5 (97.8)
Capacity Factor	0.11 (0.18)	0.09 (0.16)	0.13 (0.20)	0.10 (0.17)	0.08 (0.15)	0.12 (0.19)	0.14 (0.20)	0.11 (0.17)	0.17 (0.21)
Probability Operating	0.35 (0.48)	0.31 (0.46)	0.38 (0.49)	0.34 (0.47)	0.31 (0.46)	0.37 (0.48)	0.37 (0.48)	0.32 (0.47)	0.42 (0.49)
Probability Marginal	0.09 (0.29)	0.10 (0.30)	0.09 (0.29)	0.10 (0.30)	0.10 (0.31)	0.09 (0.29)	0.09 (0.28)	0.09 (0.28)	0.09 (0.29)
Realized Markup (\$/MWh)	-2.74 (79.84)	-9.71 (39.95)	4.24 (105.19)	-4.14 (81.35)	-11.18 (41.27)	2.91 (106.96)	2.56 (73.61)	-4.13 (33.93)	9.28 (98.04)
Resid. Demand Slope (MW per \$/MWh)	1234 (703)	1442 (663)	1064 (688)	1229 (704)	1435 (667)	1057 (689)	1253 (696)	1466 (647)	1088 (687)
Market Size (GW)	86.6 (29.5)	88.1 (28.5)	85.1 (30.4)	85.6 (30.5)	87.2 (29.6)	84.1 (31.3)	90.3 (24.8)	91.7 (23.7)	89.0 (25.8)
Panel B: Generator Characteristics									
Heat Rate (MMBtu/MWh)		10.7 (3.0)			10.9 (3.0)			10.0 (3.1)	
CO ₂ Rate (tons/MWh)		0.6 (0.2)			0.6 (0.2)			0.5 (0.2)	
NO _x Rate (lbs/MWh)		1.150 (1.967)			1.158 (1.736)			1.121 (2.686)	
SO ₂ Rate (lbs/MWh)		0.010 (0.044)			0.011 (0.049)			0.007 (0.006)	
Marginal Cost (\$/MWh)		31.73 (11.27)			32.44 (10.31)			29.03 (14.09)	
Avg. Capacity (MW)		366 (446)			335 (448)			485 (417)	
Panel C: Sample Composition									
# of Unique Generators		558			443			115	
# Simple Cycle		211			185			26	
# Combined Cycle		298			219			79	
# Steam Turbine		49			39			10	
# of Counties		50			32			18	
# in Houston Zone		220			208			12	
# in North Zone		136			62			74	
# in South Zone		152			140			12	
# in West Zone		50			33			17	

Notes. This table presents summary statistics of natural gas generators in ERCOT. Each entry reports the mean and counts of natural gas generators, with standard deviations in parentheses. Panels A and B report statistics computed over 2018–2021, the period before significant storage build-out in ERCOT. Panel C reports sample composition over the full estimation period (2018–2024). Columns (1)–(3) report statistics for all natural gas generators; Columns (4)–(6) for generators sharing the same county with storage; Columns (7)–(9) for generators located not in the same county with any storage. Panel A describes the variation in market outcome and characteristics within a day. Panel B describes the generator characteristics and marginal costs. Panel C describes sample composition.

county with storage units, while Columns (7)–(9) present statistics for generators located in different counties from any storage units. Because the operational patterns and market conditions of

these groups closely mirror the overall sample, the treatment group in my difference-in-differences design is highly representative of ERCOT natural gas generators. The LMP spread between early and late periods does not differ significantly between the treatment and control groups, alleviating concerns of endogenous storage siting relative to natural gas generators location.

3.3 Descriptive Statistics of Battery Storage Units

This section describes the characteristics, ownership, siting, and operation of battery storage units in ERCOT.

Table 2: Summary statistics of storage units

	All Storage (1)	Collocated w/ Gas (2)	First collocated w/ Gas (3)	Collocated w/ Solar (4)	Collocated w/ Wind (5)
Panel A: Storage Characteristics					
Avg. Capacity (MW)	49.3 (51.3)	51.0 (53.3)	61.4 (61.7)	61.7 (38.9)	53.0 (53.1)
Avg. Energy Capacity (MWh)	74.6 (87.7)	79.6 (94.5)	89.1 (96.5)	84.4 (43.4)	74.7 (69.0)
Avg. Duration (hr)	1.37 (0.60)	1.36 (0.64)	1.34 (0.47)	1.46 (0.72)	1.60 (0.85)
Year of First Appearance	2023.0 (1.4)	2023.1 (1.5)	2022.5 (1.6)	2022.9 (1.2)	2022.6 (1.0)
Year of First Operation	2023.1 (1.5)	2023.0 (1.4)	2022.9 (1.6)	2023.4 (1.5)	2023.8 (0.5)
Cumulative Capacity Built (GWh)	13.0	6.3	2.7	2.4	0.5
Cumulative Capacity Operational (GWh)	12.3	6.0	2.7	2.0	0.5
Panel B: Ownership Structure					
By Parent Company					
Share w/ NG Portfolio	0.05	0.07	0.07	0.04	0.14
Share w/ Wind Portfolio	0.43	0.35	0.43	0.67	1.00
Share w/ Solar Portfolio	0.50	0.47	0.57	1.00	1.00
Share Independent	0.50	0.53	0.43	0.00	0.00
Share Collocated w/ Natural Gas	0.02	0.05	0.03	0.00	0.00
Share Collocated w/ Solar	0.15	0.13	0.17	1.00	0.00
Share Collocated w/ Wind	0.04	0.00	0.00	0.00	1.00
Panel C: Sample Composition					
# of Battery Units (Operational)	165	75	30	24	7
# of Battery Units (Built)	211	105	32	26	7
# of Counties	70	32	32	16	4
# in Houston Zone	19	19	4	2	0
# in North Zone	30	16	10	14	0
# in South Zone	53	30	12	6	0
# in West Zone	63	10	4	2	7

Notes. Summary statistics of ERCOT storage units as of December 1, 2024. I report means and counts, with standard deviations in parentheses. Column (1) includes all units. Column (2) includes units in a county with a natural gas generator. Column (3) includes the first storage unit to enter each of those counties. Columns (4) and (5) include units collocated with solar PV and wind generators.

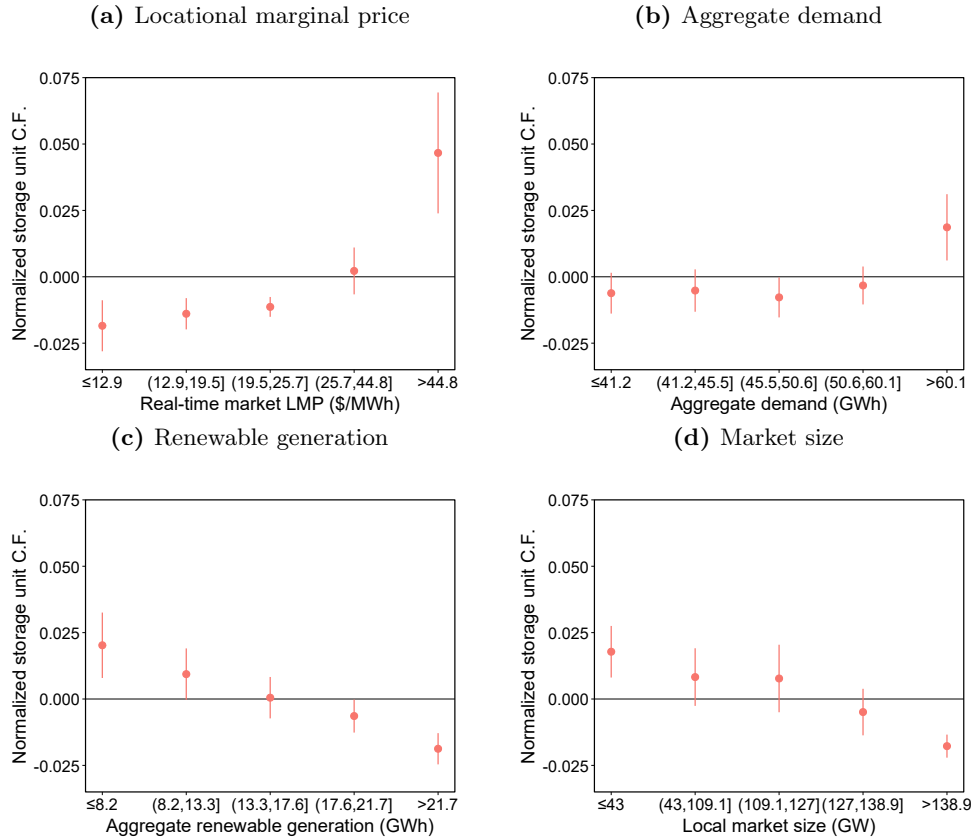
Storage in ERCOT consists mostly of short-duration batteries. Table 2, Column (1), shows

that 13 GWh of storage (211 units) is built by the end of my sample, with 12.3 GWh (165 units) operating commercially. The average unit has 49.3 MW of capacity and a 1.37-hour duration. Units are spread across the state, with the largest shares in the West and South load zones.

Storage that shares a county with a natural gas generator provides my identifying variation. Column (2) shows 6.3 GWh of such capacity (106 units), with 6 GWh (75 units) operating commercially. Capacity and duration resemble the full sample. Because gas generators cluster near population centers, these units sit more often in the Houston and South zones and less often in the West. Column (3) shows that the first entrant into each county varies widely in capacity.

Ownership and collocation differ across storage units. Panel B shows that about half of the units belong to storage-only firms, while the rest belong to firms that also own renewables. Storage often shares a node with solar PV, mainly in the North zone, and rarely with wind, which is entirely in the West. Storage and gas units rarely share a node.

Figure 2: Battery storage capacity factor by market conditions



Notes: Each panel shows the conditional mean and 95% confidence interval of storage capacity factor by quantile of the explanatory variable. Capacity factor is energy charged or discharged divided by the maximum possible in each interval; positive values indicate charging and negative values indicate discharging. Market size is total installed capacity in the local market. Following [Lamp and Samano \(2022\)](#), I estimate $\text{StorageCF}_{it} = \sum_q \beta_q \mathbb{1}(X_{it} \in q\text{-th quantile of } \mathbf{X}) + \varepsilon_{it}$ for generator i in interval t . LMP and market size vary by generator and interval; demand and renewable generation vary by interval. Standard errors are two-way clustered by generator and day.

Storage operation appears consistent with price arbitrage. Figure 2 shows that units tend

to charge when LMPs are low, demand is low, and renewable output is high. Units also tend to charge during less congested periods (larger local markets) and discharge during congested hours. [Woerman \(2023\)](#) documents that firms exercise more market power in smaller markets. This pattern suggests that storage entry may affect competition with gas generators where market power is greatest.

4 Theoretical Framework

I build a simple two-period model to show how storage entry changes the competitive behavior of natural gas generators (gas generators, hereafter). Storage affects a gas generator's markup through two channels. It shifts the *level* of the residual demand the generator faces, which changes the quantity it serves. It also changes the *slope* of that residual demand, which changes how easily the rest of the market substitutes away from it. The appendix extends the model to transmission congestion, strategic storage, and portfolio ownership.

4.1 Setup

The model has two periods, $t \in \{L, H\}$, with inelastic load $D_L < D_H$, and each period clears at a single locational price p_t . A competitive fringe of price-taking units has supply $S_t^c(p) = \alpha_t + \gamma p$ with $\gamma > 0$ ([Fabra and Imelda, 2023](#); [Ito and Reguant, 2016](#)). A storage fleet charges in the low-demand period ($s_L \leq 0$) and discharges in the high-demand period ($s_H \geq 0$). A strategic gas firm f with marginal cost c_{ft} chooses output q_{ft} to maximize $\pi_{ft} = (p_t - c_{ft}) q_{ft}$, taking residual demand as given. The market clears in each period,

$$(1) \quad S_t^c(p_t) + q_{ft} + s_t(p_t) = D_t.$$

Storage supply. Each battery maximizes arbitrage profit. To be able to sell one MWh in the discharge period, a battery with round-trip efficiency η must buy $1/\eta$ MWh in the charge period, so its marginal cost is $m_s = p_L/\eta$. Because the objective is linear in the arbitrage quantity, it discharges at full capacity when $p_H > m_s$ and stays idle otherwise. A single battery's offer is a step, consistent with awarded storage quantities clustering on the vertical segment of the offer curve observed in the storage bid data ([Figure E.20](#)). Batteries differ in efficiency, state of charge, and price expectations, so a higher price pulls successively more of them past their threshold, and the aggregate supply is smooth ([Karaduman, 2023](#)),

$$(2) \quad s_t(p_t) = \bar{s}_t + \sigma_t p_t, \quad \sigma_t \geq 0.$$

The level $\bar{s}_t$ is storage that clears at any relevant price. It is positive when storage discharges and negative when it charges. The slope σ_t is storage near its threshold, which responds to the price in both periods.

4.2 The Generator's Problem

The gas firm's optimal markup is proportional to the quantity it serves and inversely proportional to the slope of its residual demand. Substituting fringe supply and storage into Equation (1), the firm faces residual demand

$$(3) \quad q_{ft} = A_t - E_t p_t, \quad A_t = D_t - \alpha_t - \bar{s}_t, \quad E_t = \gamma + \sigma_t.$$

The first-order condition gives the markup rule

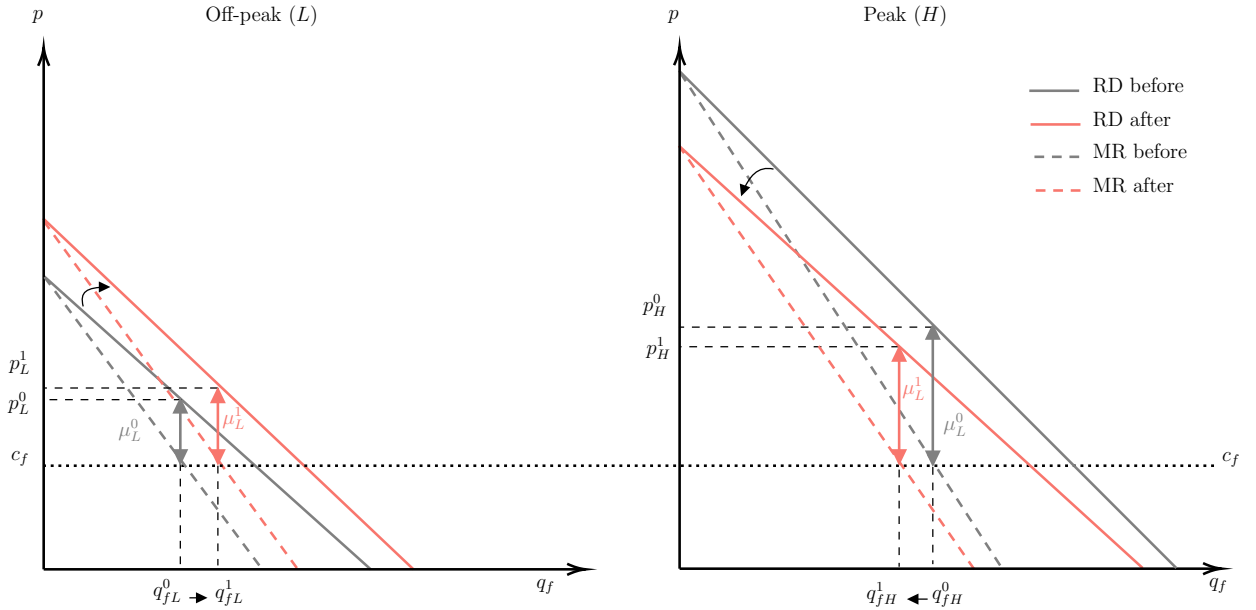
$$(4) \quad \mu_{ft} \equiv p_t - c_{ft} = \frac{q_{ft}}{E_t},$$

standard in the electricity market literature (Jha and Leslie, 2025; Woerman, 2023; Ryan, 2021; Doshi, 2024; Fabra and Imelda, 2023). Solving jointly with residual demand gives the equilibrium

$$(5) \quad p_t^* = \frac{1}{2} \left(\frac{A_t}{E_t} + c_{ft} \right), \quad q_{ft}^* = \frac{1}{2} (A_t - E_t c_{ft}), \quad \mu_t^* = \frac{A_t}{2E_t} - \frac{c_{ft}}{2}.$$

The markup rises with the level of residual demand and falls with its slope. Figure 3 illustrates both channels. I abstract from any effect of storage on the generator's own marginal cost.

Figure 3: Illustration of market power effect of storage entry



Notes: Residual demand of a natural gas generator before (gray) and after (red) storage entry, in the charge (left) and discharge (right) periods. Setting $MR = c$ yields the markup μ . Storage entry shifts residual demand outward in the charge period and inward in the discharge period, and makes it flatter in both.

4.3 Storage Entry and the Generator's Outcomes

Local storage shifts both the level and slope of residual demand. Let storage capacity, B , be the installed battery energy capacity in the generator's local market, thus,

$$(6) \quad \frac{\partial A_H}{\partial B} \leq 0, \quad \frac{\partial A_L}{\partial B} \geq 0, \quad \frac{\partial E_t}{\partial B} = \frac{\partial \sigma_t}{\partial B} \geq 0.$$

In this simple model, storage lowers the level of residual demand during the discharge period, raises it when it charges, and flattens it in both periods. The following comparative statics describe how storage affects the generator's output and markup.

Generation. Differentiating the equilibrium output with respect to B ,

$$(7) \quad \frac{\partial q_{ft}}{\partial B} = \frac{1}{2} \left(\frac{\partial A_t}{\partial B} - c_{ft} \frac{\partial E_t}{\partial B} \right), \quad \frac{\partial^2 q_{ft}}{\partial B \partial c_{ft}} = -\frac{1}{2} \frac{\partial E_t}{\partial B} \leq 0.$$

In the discharge period, both terms lower the gas generator's output. In the charge period, the added charging load raises output while the flatter residual demand lowers it, so the sign is an empirical question. The cross-derivative implies that storage displaces higher-cost generators more, because a flatter residual demand leaves less room for units whose cost is close to the price.

Markup. Differentiating the markup rule (4) separates the two channels,

$$(8) \quad \frac{\partial \mu}{\partial B} = \underbrace{\frac{1}{E} \frac{\partial q}{\partial B}}_{\text{displacement}} - \underbrace{\frac{q}{E^2} \frac{\partial E}{\partial B}}_{\text{slope}}.$$

The slope term is always negative because a more elastic residual demand raises the cost of withholding. In the discharge period, both terms are negative, so the markup falls. In the charge period, the two terms oppose, and the markup rises only if storage raises the quantity served proportionally more than it flattens residual demand, $\frac{1}{q} \frac{\partial q}{\partial B} > \frac{1}{E} \frac{\partial E}{\partial B}$.

4.4 Summary of Theoretical Results

The model produces three theoretical predictions that motivate the empirical analysis. First, storage flattens the residual demand of gas generators in both periods. Second, in the discharge period, storage lowers both the output and the markup of gas generators, with higher-cost generators losing more output. Third, in the charge period, storage increases load but still flattens residual demand, so the net effects on output and markup are ambiguous, and the markup can rise only if output rises.

5 Empirical Strategy

I take the model to the data using a panel of marginal gas generators in ERCOT at the hourly level. I estimate the marginal effect of storage separately by hour of the day, with q_{iht} , μ_{iht} , and c_{it} the output, markup, and marginal cost of generator i on date t at hour h . Since storage production patterns exhibit a daily cycle, I group hours into charging periods: overnight (midnight to 6 AM), midday (10 AM to 4 PM); and discharging periods: the evening peak (5 PM to 9 PM). I normalize output as a capacity factor, $y_{iht} = q_{iht}/K_i$, where K_i is the capacity of generator i to account for large variation in gas generators capacity in ERCOT. The capacity factor puts every generator on a common scale, and multiplying by K_i recovers the output in MWh.

I employ two research designs. I begin with estimating the optimal output rule from the theoretical model by the instrumental variable approach. The parameter estimates describe how generator output responds to its own cost as the model requires, and motivate the interaction between storage and cost that informs the second design. I then employ a continuous two-stage difference-in-differences design as the main design for all outcomes: output, operating states, the residual demand slope, markups, and operating profit.

5.1 Displacement of Gas Generation

I bring the theoretical prediction from the previous section to the data. The generator's optimal output in Equation (5) is linear in its marginal cost, $q^* = \frac{1}{2}(A - E c)$. Dividing by capacity keeps the rule linear, $y^* = \frac{1}{2}(\tilde{A} - \tilde{E}c)$ with $\tilde{A} = A/K$ and $\tilde{E} = E/K$. I assume local storage shifts the intercept and slope linearly at hour h , $\tilde{A}_h(B) = a_h + a'_h B$ and $\tilde{E}_h(B) = e_h + e'_h B$. Substituting and writing $c = \bar{c}_k + \tilde{c}$, with $\bar{c}_k$ the mean marginal cost of technology k , gives

$$(9) \quad y_h^* = \frac{1}{2}(a_h - e_h \bar{c}_k) + \frac{1}{2}(a'_h - e'_h \bar{c}_k) B - \frac{1}{2}e_h \tilde{c} - \frac{1}{2}e'_h B \tilde{c}.$$

The coefficient on B is the displacement at the typical cost of the technology, and the coefficient on $B\tilde{c}$ shows how displacement varies with cost. Because $e'_h \geq 0$ in the model, higher-cost generators are displaced more.

Estimating equation. I estimate the empirical analog of Equation (9) on interior hours,

$$(10) \quad y_{iht} = \delta^{(1)} B_{m(i),t} + \delta^{(2)} \tilde{c}_{it} + \delta^{(3)} (B_{m(i),t} \tilde{c}_{it}) + \mathbf{X}_{iht} \theta + \alpha_i + \nu_{\ell(i),y(t),h} + u_{iht}.$$

The outcome y_{iht} is the capacity factor in percentage points. The treatment $B_{m(i),t}$ is the cumulative storage energy capacity in the generator's county $m(i)$, in units of 100 MWh. The cost term $\tilde{c}_{it} = c_{it} - \bar{c}_k$ is marginal cost in \$/MWh, centered at the mean of its technology on interior hours. The generator fixed effect α_i absorbs permanent differences across generators, and the load-zone-by-month-by-hour fixed effect $\nu_{\ell(i),y(t),h}$ absorbs regional shifts in load and fringe supply. The controls $\mathbf{X}_{iht}$ are system net load, a cubic in local temperature, and local renewable capacity and

potential output: county solar and wind capacity, and their products with local irradiance and a normalized wind power curve. I cluster standard errors by county. I estimate the equation pooled, by technology $k \in \{\text{CC}, \text{SC}, \text{GS}\}$, and by technology and hour. Because some estimates rely on as few as 18 county clusters, I construct confidence intervals with critical values from a t distribution with $G - 1$ degrees of freedom, where G is the number of counties behind each estimate (Bester, Conley and Hansen, 2011; Colin Cameron and Miller, 2015).

Interior hours are hours in which the generator runs strictly between its low and high sustained limits. The first-order condition in Equation (4) holds with equality only in the interior. At the low sustained limit, a committed generator’s output reflects the startup and minimum-load costs of staying online (Jha and Leslie, 2025), and at the high sustained limit, capacity binds.

Instruments. Marginal cost is potentially endogenous. Realized heat rates depend on realized output, so the constructed cost is mechanically a function of output (Demirer and Karaduman, 2026). The constructed cost does not factor in maintenance, startup, and ramping costs, so it is a noisy proxy. The identity of the marginal unit might respond to the shocks that enter u_{iht} .¹⁴ I instrument $\tilde{c}_{it}$ and $B_{m(i),t}\tilde{c}_{it}$ with $\tilde{z}_{it}$ and $\tilde{z}_{it}B_{m(i),t}$, where $z_{it} = \text{HR}_i^{\text{pre}} \times P_t^{\text{gas}}$ is the product of the generator’s pre-entry heat rate and the Henry Hub gas price, centered within technology. The instrument shifts cost strongly, because marginal cost is dominated by fuel cost which by construction is heat rate times gas price. The pre-entry heat rate does not move with the current operating point, which breaks the mechanical link between cost and output.

Identifying assumptions. Identification rests on four assumptions. Let W_{iht} collect the fixed effects and controls. The instruments must satisfy relevance and exclusion restriction, $\mathbb{E}[\tilde{z}_{it}u_{iht} | W_{iht}] = 0$ and $\mathbb{E}[\tilde{z}_{it}B_{m(i),t}u_{iht} | W_{iht}] = 0$. Storage must be conditionally exogenous, $\mathbb{E}[B_{m(i),t}u_{iht} | W_{iht}] = 0$, and the generator must optimize in the interior.

There are several potential threats to these identification assumptions. First, the Henry Hub price shifts the costs of all gas generators at once and may therefore shift residual demand directly; replacing the month fixed effects with load-zone-by-date-by-hour fixed effects absorbs this common shock. Second, storage enters the estimating equation as an exogenous regressor, so its coefficient is exposed to comparisons with already-treated counties and to siting on local trends; the two-stage design below addresses both. Third, storage can also change which generators operate in the interior, so I report how sensitive the estimate is to excluding hours near the operating limits and estimate the effect on operating states directly. Finally, the coefficients in Equation (9) equal the residual demand slope only under constant marginal cost. With increasing marginal cost, $MC = c + kq$, the cost coefficient identifies $E/(2 + Ek)$, which approaches $1/k$ when residual demand is flat. I therefore use this specification to test for the sign of the cost interaction and measure the slope directly.

¹⁴Ganapati, Shapiro and Walker (2020) instrument estimated marginal costs with energy-price shocks to address measurement error, and Fabra and Reguant (2014) treat hourly marginal cost as endogenous because the identity of the marginal unit is set by unobserved supply and demand shocks.

5.2 The Two-Stage Difference-in-Differences Design

Treatment. The treatment is the entry of storage into a generator’s county. I define the commercial operation date $\text{COD}_{m(i)}$ as the first date a same-county storage unit is awarded a positive quantity in the real-time market. The treatment is the continuous dose $B_{m(i),t}$, and $\mathbb{1}\{B_{m(i),t} > 0\}$ is the entry indicator. The county boundary approximates the competitive neighborhood that transmission constraints create (Figure E.22).

Estimating equation. For an hourly outcome Y_{iht} , the estimating equation is

$$(11) \quad Y_{iht} = \sum_{h=0}^{23} \beta_h \mathbb{1}\{\text{hour} = h\} B_{m(i),t} + \alpha_i + \eta_{\ell(i),t,h} + \mathbf{X}_{iht}\theta + \varepsilon_{iht},$$

where $\eta_{\ell(i),t,h}$ is a load-zone-by-date-by-hour fixed effect and $\mathbf{X}_{iht}$ contains the same controls as in Equation (10). For the capacity factor, I carry the cost gradient from the structural model into this design by adding an interaction with predetermined fuel cost,

$$(12) \quad y_{iht} = \sum_{h=0}^{23} \mathbb{1}\{\text{hour} = h\} (\beta_h B_{m(i),t} + \gamma_h \tilde{z}_{it} B_{m(i),t}) + \lambda \tilde{z}_{it} + \alpha_i + \eta_{\ell(i),t,h} + \mathbf{X}_{iht}\theta + \varepsilon_{iht}.$$

The coefficient β_h is the displacement at mean predetermined cost, and γ_h is the reduced-form analog of $\delta^{(3)}$. I use predetermined cost rather than realized cost because realized cost responds to storage. I cluster standard errors at the county level.

Estimation. Counties adopt storage at different dates and the dose grows over time. Under this staggered timing, the two-way fixed-effects estimator uses already-treated counties as controls, which can attenuate or reverse the sign of the true effect when effects are heterogeneous (Goodman-Bacon, 2021; De Chaisemartin and D’Haultfoeuille, 2020; Callaway and Sant’Anna, 2021; Sun and Abraham, 2021; Borusyak, Jaravel and Spiess, 2024). I therefore use the two-stage estimator of Gardner et al. (2024). The first stage estimates the fixed effects and control coefficients, including λ , on never-treated and not-yet-treated observations. The second stage regresses the first-stage residuals on the storage terms. The standard errors account for the first stage.

With a continuous and time-varying dose, β_h is a positively weighted average of the local slopes of the dose-response function (Yitzhaki, 1996; Gardner et al., 2024),

$$(13) \quad \hat{\beta}_h = \mathbb{E} \left[\sum_d w(d) \frac{\partial \mathbb{E}[Y_{iht}(d) - Y_{iht}(0) \mid B_{m(i),t} = d]}{\partial d} \right],$$

where the weights $w(d) \geq 0$ reflect the distribution of the dose among treated generators. The coefficient is therefore a marginal effect scaled to per 100 MWh rather than the effect of moving from no storage to 100 MWh. The identification results of Callaway, Goodman-Bacon and Sant’Anna (2024) assume a dose fixed at a single value per unit, which does not hold in this setting.

Identifying assumptions. Identification rests on three assumptions. The first is parallel trends: absent local entry, the untreated outcome is additive in the fixed effects and controls, $\mathbb{E}[Y_{iht}(0) \mid \alpha_i, \eta_{\ell(i),t,h}, \mathbf{X}_{iht}] = \alpha_i + \eta_{\ell(i),t,h} + \mathbf{X}_{iht}\theta$. The second is no anticipation, $Y_{iht}(d) = Y_{iht}(0)$ for $t < \text{COD}_{m(i)}$. The third is no interference, so a generator’s outcome depends only on its own county’s dose.

I test parallel trends with an event study that replaces the dose with indicators for years relative to entry,

$$(14) \quad Y_{iht} = \sum_{\substack{k=-3 \\ k \neq -1}}^3 \tau_k \mathbb{1}\{K_{m(i),t} = k\} + \alpha_i + \eta_{\ell(i),t,h} + \mathbf{X}_{iht}\theta + \varepsilon_{iht},$$

where $K_{m(i),t}$ counts years relative to entry, binned up to three years pre and post, and the year before entry is the reference. I estimate it by hour group and at the daily level for operating profit. Anticipation is unlikely, because storage appears in the SCED data long before commercial operation, with a median gap of 204 days (Figure E.23), and the data are published with a sixty-day lag. The remaining concern is that storage enters where local conditions are changing. The generator and load-zone-by-date-by-hour fixed effects absorb permanent and regional differences, and the event study then provides a falsification test for county-specific trends. Storage is also often sited alongside new solar capacity, whose output displaces gas generation directly. To address this I control for the local renewable capacity and potential.

Outcomes. For output, the outcome is the capacity factor, including hours with zero output, estimated by technology and weighted by capacity. Weighting makes β_h a capacity-weighted average of generator-level effects, so multiplying it by capacity and summing recovers the change in MWh. This is the displacement I carry into the welfare analysis. I also estimate the effect on indicators for running, running at minimum load, running in the interior, and running at maximum output, which show how storage changes the way generators operate. For the residual demand slope, the outcome E_{iht} is the slope of the local residual demand curve constructed from the offer stack.¹⁵ For the markup, the outcome is $\mu_{iht} = \text{LMP}_{iht} - c_{it}$, set to zero when the generator does not run. For operating profit, I aggregate to the generator-day and use daily operating profit per MW of capacity, $\pi_{it} = \sum_h \mu_{iht} q_{iht} / K_i$, weighted by capacity, with load-zone-by-date fixed effects,

$$(15) \quad \pi_{it} = \beta^\pi B_{m(i),t} + \alpha_i + \eta_{\ell(i),t} + \mathbf{X}_{it}\theta + \varepsilon_{it}.$$

I also estimate (15) separately for the competitive rent and the market power rent defined in Section 6.2. Because the estimator is linear, the two effects sum to β^π .

¹⁵Section 3 details the construction of E_{iht} . A larger value denotes a flatter, more elastic residual demand.

6 Results

6.1 Displacement results

This section presents the estimates on output and the slope of the residual demand of gas generators. First, I compare the structural estimates of Equation (10) with the two-stage difference-in-differences (2SDD) estimates. I then present the impact on hourly output by technology, the operating margins, and the residual demand slope. I then estimate the aggregate displacement that informs the welfare analysis.

Table 3: Local storage effect on capacity factor: structural and two-stage estimates

	Capacity factor (pp)			
	OLS (1)	IV (2)	2SDD (3)	2SDD (4)
Storage B (100MWh)	-0.791*** (0.291)	-0.792*** (0.292)	-0.407*** (0.134)	-0.906*** (0.267)
Cost $\tilde{c}$ (\$/MWh)	-0.067*** (0.017)	-0.068*** (0.016)		
Cost $\tilde{c} \times B$	-0.016*** (0.003)	-0.016*** (0.003)		
Predetermined cost $\tilde{z} \times B$			-0.016*** (0.004)	-0.004 (0.008)
Generator FE	✓	✓	✓	✓
Load zone-Month-Hour FE	✓	✓		
Load zone-Date-Hour FE			✓	✓
Local renewable capacity and potential	✓	✓	✓	✓
Local temperature	✓	✓	✓	✓
Sample	Interior	Interior	Interior	All hours
Weights	None	None	None	Capacity
Observations	3,808,488	3,808,488	3,690,817	13,575,368
Counties	50	50	50	50
First-stage F , $\tilde{c}$		332.0		
First-stage F , $\tilde{c} \times B$		2429.5		
Partial R^2 , $\tilde{c}$		0.385		
Partial R^2 , $\tilde{c} \times B$		0.810		

Notes. The outcome is the hourly capacity factor in percentage points. B is cumulative storage energy capacity in the generator's county in 100 MWh. Columns (1) and (2) estimate Equation (10), with marginal cost centered within technology and instrumented in Column (2) by the pre-entry heat rate times the Henry Hub price. Columns (3) and (4) use the two-stage difference-in-differences estimator, with storage interacted with predetermined cost, the pre-entry heat rate times the Henry Hub price, centered within technology. Interior hours have output strictly between the low and high sustained limits. Standard errors clustered by county in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 3 presents the pooled estimates. Column (1) and Column (2) present the OLS and IV estimates of Equation (10) on interior hours. An additional 100 MWh of local storage lowers the capacity factor of a generator with an average marginal cost by 0.79 percentage points. The OLS and IV estimates are nearly identical, which suggests that simultaneity in the constructed marginal cost is modest. The instruments are highly relevant, yielding first-stage cluster-robust F -statistics of 332 for cost and 2,430 for the interaction, with partial R^2 of 0.39 and 0.81. The cost coefficient is

negative consistent with the optimal output rule. The interaction term is also negative, so storage displaces higher-cost generators more.

Column (3) applies the two-stage estimator to the same interior sample. The storage coefficient is -0.41 (s.e. 0.13), about half the IV estimate, although the confidence intervals overlap. The two-stage estimator uses only generators in not-yet-treated counties as controls, so it avoids the comparisons with already-treated generators that affect the storage coefficient in Equation (10). The interaction with predetermined cost shifter is -0.016 (s.e. 0.004), the same sign and magnitude as the structural interaction. This implies that higher-cost generators are displaced more under both designs. Column (4) is my preferred specification for the welfare analysis. It includes all hours, including hours with zero output and hours at an operating limit, and weights by capacity. Storage lowers the capacity factor by 0.91 percentage points (s.e. 0.27).

The two-stage estimate is more robust to the sample of operating hours. Appendix Table C.1 shows that excluding hours within 2 percent of capacity of either operating limit shrinks the IV estimate to -0.26 and makes it insignificant. The same restriction leaves the two-stage estimate at -0.46 and significant (Appendix Table C.3). Much of the IV estimate is driven by generators operating close to their limits. The IV estimate is otherwise stable across choice of fixed effect and sample weights. It is -0.81 when I include hours at the operating limits, -0.66 with capacity weights, and -0.82 with load-zone-by-date-by-hour fixed effects that absorb the common natural gas price shock.

Table 4 presents the welfare specification by technology. Storage lowers the capacity factor of combined cycle generators by 0.40 percentage points (s.e. 0.19), or about 3.7 MWh at their mean capacity of 913 MW. The estimates for simple cycle (-0.93) and steam turbine (-0.20) generators are not statistically significant. The pooled estimate of -0.91 exceeds the combined cycle and steam turbine estimates because the pooled first stage shares fixed effects across technologies and so also compares generators of different types. For aggregation, I therefore use the technology-specific estimates.

Hourly generation displacement Storage displaces gas generation in the afternoon and evening. Figure 4 presents the hourly two-stage estimates by technology. Combined cycle output falls significantly from 1 PM to 10 PM, with the largest effect of 0.82 percentage points at 7 PM, or about 7.5 MWh at mean capacity. Simple cycle output falls significantly from 1 PM to 9 PM, with the largest effect of 2.05 percentage points at 4 PM, or about 2.7 MWh at mean capacity. None of the overnight estimates is significant, and none of the steam turbine estimates is significant in any hour. The confidence intervals use t_{G-1} critical values, where G is the number of counties behind each estimate.

The theoretical model predicts that storage displaces higher-cost generators more. Panel (b) of Figure 4 evaluates the displacement at the 25th and 75th percentiles of predetermined cost within each technology. Consistent with the prediction, the higher-cost simple cycle generator loses 0.6 to 0.8 percentage points more capacity factor than the lower-cost one from 1 PM to 8 PM, and the

Table 4: Local storage effect on capacity factor by technology

	Capacity factor (pp)			
	All gas (1)	Combined Cycle (2)	Simple Cycle (3)	Steam Turbine (4)
Storage B (100MWh)	-0.906*** (0.267)	-0.403** (0.185)	-0.926 (0.570)	-0.203 (0.466)
Predetermined cost $\tilde{z} \times B$	-0.004 (0.008)	-0.014 (0.012)	-0.007 (0.009)	0.014 (0.012)
Generator FE	✓	✓	✓	✓
Load zone-Date-Hour FE	✓	✓	✓	✓
Local renewable capacity and potential	✓	✓	✓	✓
Local temperature	✓	✓	✓	✓
Sample	All hours	All hours	All hours	All hours
Weights	Capacity	Capacity	Capacity	Capacity
Observations	13,575,368	3,527,926	7,215,427	2,410,334
Counties	50	33	25	18

Notes. Two-stage difference-in-differences estimates on all hours, weighted by nameplate capacity. Column (1) pools all gas generators and repeats Column (4) of Table 3. Standard errors clustered by county in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

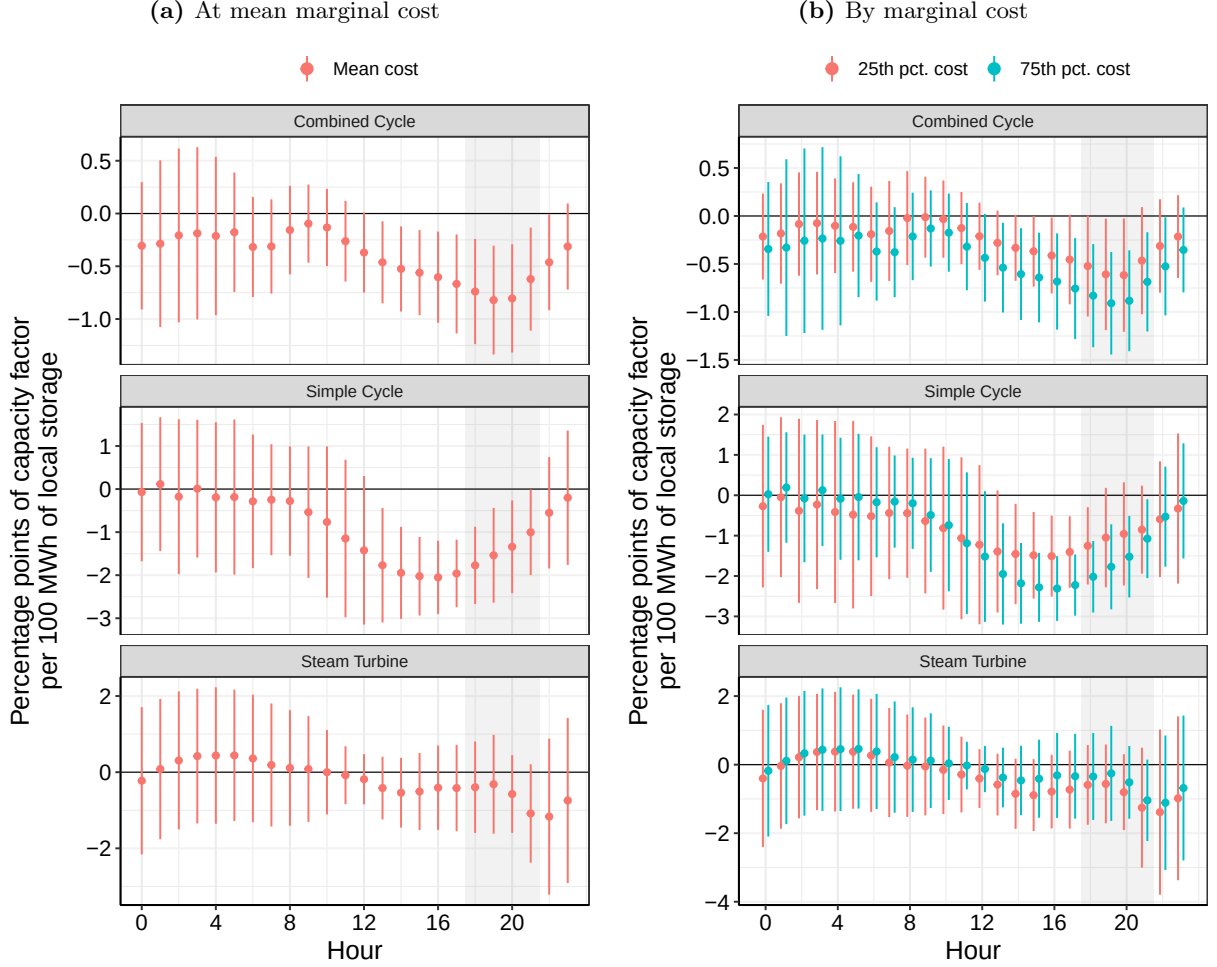
cost interaction is significant in these hours. The higher-cost combined cycle generator loses 0.2 to 0.3 percentage points more in every hour, although the interaction is not significant hour by hour. Steam turbines show the opposite gradient, and it is not significant.

I perform two robustness checks that support a causal reading of the afternoon displacement. First, the local renewable controls matter little for the afternoon estimates (Appendix Figure C.7). They lower the combined cycle estimates by about 0.1 percentage points in every hour and remove most of the overnight simple cycle displacement. The afternoon simple cycle displacement remains unchanged, at about 2.0 percentage points at 4 PM. This implies that the afternoon displacement is unlikely to be driven by solar capacity that enters alongside storage. Second, the displacement is larger in the Houston and North load zones, where it reaches 1.6 percentage points in the evening, than in the wind-rich West and South zones. It is also larger on days with high solar or wind output, when arbitrage is more valuable (Appendix Figure C.8).

Operating margins Table 5 presents estimates on operating margin. For combined cycle, the evening displacement works through fewer hours at maximum output, which fall by 1.6 percentage points. For simple cycle, storage lowers the probability of running at minimum load during midday by 1.6 percentage points. Steam turbines behave differently. In the evening peak, storage increases the probability that a steam turbine runs by 1.6 percentage points and runs in the interior by 2.6 percentage points, while lowering the probability of running at maximum output by 1.6 percentage points. Steam turbines run more but further from their maximum output.

Residual demand slope The model predicts that storage makes the residual demand of gas generators flatter. To test this prediction, I use the slope of the local residual demand constructed

Figure 4: Hourly displacement by technology



Notes. Two-stage difference-in-differences estimates of the effect of 100 MWh of local storage on the capacity factor, in percentage points, by hour and technology, on all hours and weighted by capacity. Panel (a) evaluates the effect at the mean predetermined cost of each technology. Panel (b) evaluates it at the 25th and 75th percentiles of predetermined cost. The 95% confidence intervals use t_{G-1} critical values and standard errors clustered at the county level. The shaded band marks the evening peak.

from the bid data as the outcome in the two-stage design. Before storage entry, the slope is about 1,300 MW per \$/MWh overnight and about 750 to 900 MW per \$/MWh in the afternoon and evening. Figure 5 shows that each 100 MWh of local storage increases the slope by 1.4 to 4.4 MW per \$/MWh, or 0.2 to 0.5 percent of the baseline. The effect is statistically significant overnight and in the morning, and positive but imprecisely estimated in the evening. The event study estimates exhibit no pre-trend in any hour group (Appendix Figure C.10). Across technology, the change in slope is stronger for combined cycle generators in the evening, when the model predicts market power binds. Their residual demand becomes flatter by 5.3 to 8.0 MW per \$/MWh per 100 MWh of storage between 6 PM and 11 PM (Appendix Figure C.9).

Both the inferred residual demand slope from IV estimates and two-stage estimates on the

Table 5: Effect of local storage on operating margin

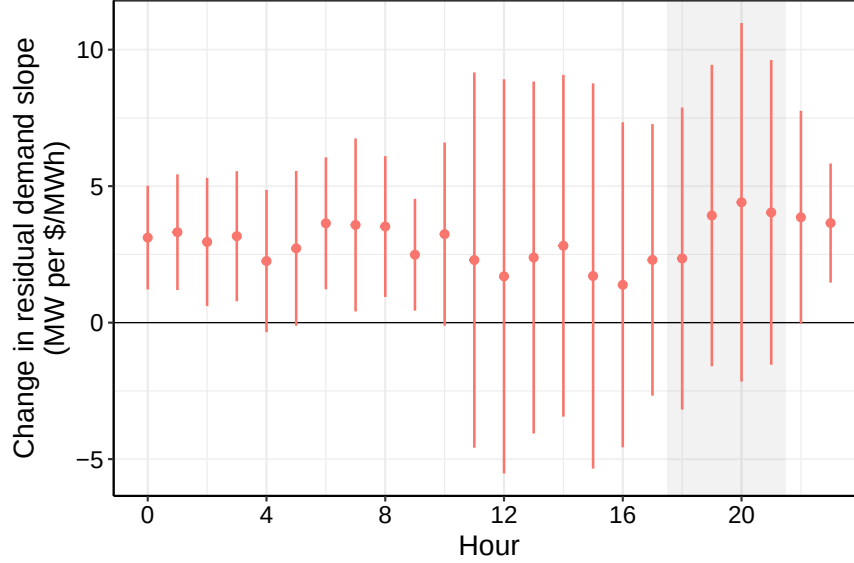
	Running	At minimum load	Interior	At maximum output
<i>Combined Cycle</i>				
Overnight (0–6)	-0.467 (0.361)	-0.420 (0.519)	-0.626 (0.792)	0.579 (0.544)
Midday (10–16)	-0.218 (0.265)	-0.088 (0.323)	0.413 (0.572)	-0.543 (0.502)
Evening (17–21)	-0.446 (0.271)	0.255 (0.324)	0.896 (0.693)	-1.597** (0.700)
<i>Simple Cycle</i>				
Overnight (0–6)	2.656 (3.142)	-0.901 (0.570)	3.565 (3.641)	-0.008 (0.242)
Midday (10–16)	1.945 (2.062)	-1.590** (0.759)	3.289 (2.602)	0.247 (0.422)
Evening (17–21)	1.611* (0.835)	-1.512 (1.039)	2.476 (2.268)	0.647 (1.163)
<i>Steam Turbine</i>				
Overnight (0–6)	0.983 (0.629)	-0.234 (1.067)	2.002** (0.802)	-0.786* (0.375)
Midday (10–16)	1.582* (0.846)	-0.033 (0.657)	1.622* (0.840)	-0.006 (0.344)
Evening (17–21)	1.647*** (0.500)	0.652 (0.629)	2.592*** (0.768)	-1.597*** (0.339)

Notes. Two-stage difference-in-differences estimates of the effect of 100 MWh of local storage on indicators for running, running at minimum load, running in the interior, and running at maximum output, in percentage points, on hours with both operating limits observed. The last three columns sum to the first. Standard errors clustered by county in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

measured slope agree in sign. The negative interaction implies a flatter residual demand for combined cycle and simple cycle generators. The measured slope agrees for combined cycle generators overnight and in the evening, where the effect of 7.4 MW per \$/MWh is significant, and for simple cycle generators at midday and in the evening. In magnitude, the slope implied by the cost coefficient is about 1.7 MW per \$/MWh for a combined cycle generator at mean capacity, roughly three orders of magnitude below the measured slope of 1,300 MW per \$/MWh. The static model with constant marginal cost cannot explain this discrepancy. With increasing marginal cost, $MC = c + kq$, the optimal output becomes $q^* = (A - Ec)/(2 + Ek)$, and when residual demand is flat the cost response approaches $-1/k$. The cost coefficient then reflects the curvature of the generator's own cost, factoring in heat rate curvature and ramping and minimum load constraints, rather than the residual demand alone. Therefore, I use the measured slope for the slope channel and interpret the structural interaction only for its sign.

Aggregate displacement Local storage displaced about 872 GWh of gas generation per year over the sample (s.e. 347). The counterfactual uses the unit-specific effect from the welfare specification, which lets displacement vary with each generator's predetermined cost. Of the total,

Figure 5: Effect of local storage on the residual demand slope



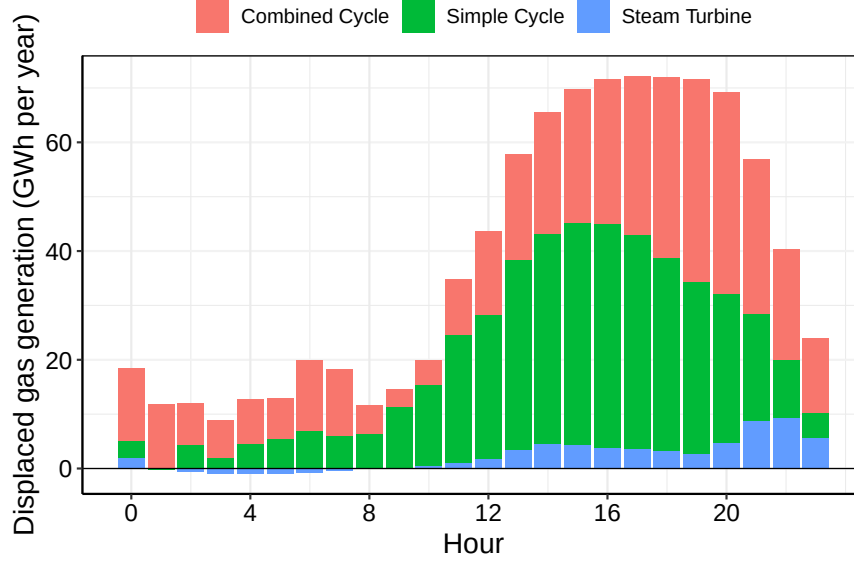
Notes. Two-stage difference-in-differences estimates of the effect of 100 MWh of local storage on the slope of the local residual demand curve, in MW per \$/MWh, by hour, for all gas generators. A larger value is a flatter, more elastic residual demand. The slope is winsorized at the 99th percentile within technology. The 95% confidence intervals use t_{G-1} critical values and standard errors clustered at the county level.

413 GWh came from combined cycle generators and 427 GWh from simple cycle generators, which are 0.28 and 2.81 percent of their observed output. Each is significant at the 10 percent level, and the total is significant at the 5 percent level. The steam turbine estimate of 32 GWh is not distinguishable from zero, and a quarter of the steam turbine counterfactuals hit the zero-output bound, so I do not interpret it. Figure 6 shows that the displaced generation is concentrated in the afternoon and evening. Average installed storage in counties with gas generation was 1,184 MWh, so the estimates imply about 74 GWh of displaced gas generation per 100 MWh of storage per year.

The point estimate is large relative to what storage itself supplies. Storage in counties with gas generation discharged 196 GWh per year, about 0.45 full cycles per day, so the estimated displacement is about four times its discharge. The lower bound of the 95 percent confidence interval, about 190 GWh per year, is close to that discharge. Two mechanisms can explain this. First, storage might relieve local congestion and allow imported energy to replace local gas generation. Gas generation might also shift to counties without storage, which the county-level design cannot separate from a true reduction. I therefore treat the aggregate as an upper bound on the storage-induced gas generation displacement. The welfare analysis in Section 7 values storage by its observed operations rather than by this displacement for the same reason.

Summary of displacement results Overall, both empirical strategies provide consistent evidence that battery storage displaces natural gas generation during peak hours. The structural

Figure 6: Gas generation displaced by local storage, by hour



Notes. Annual displaced generation, in GWh, from the technology-specific two-stage estimates by hour, on all hours and weighted by capacity. Counterfactual output is observed output net of the estimated storage contribution, bounded between zero and capacity.

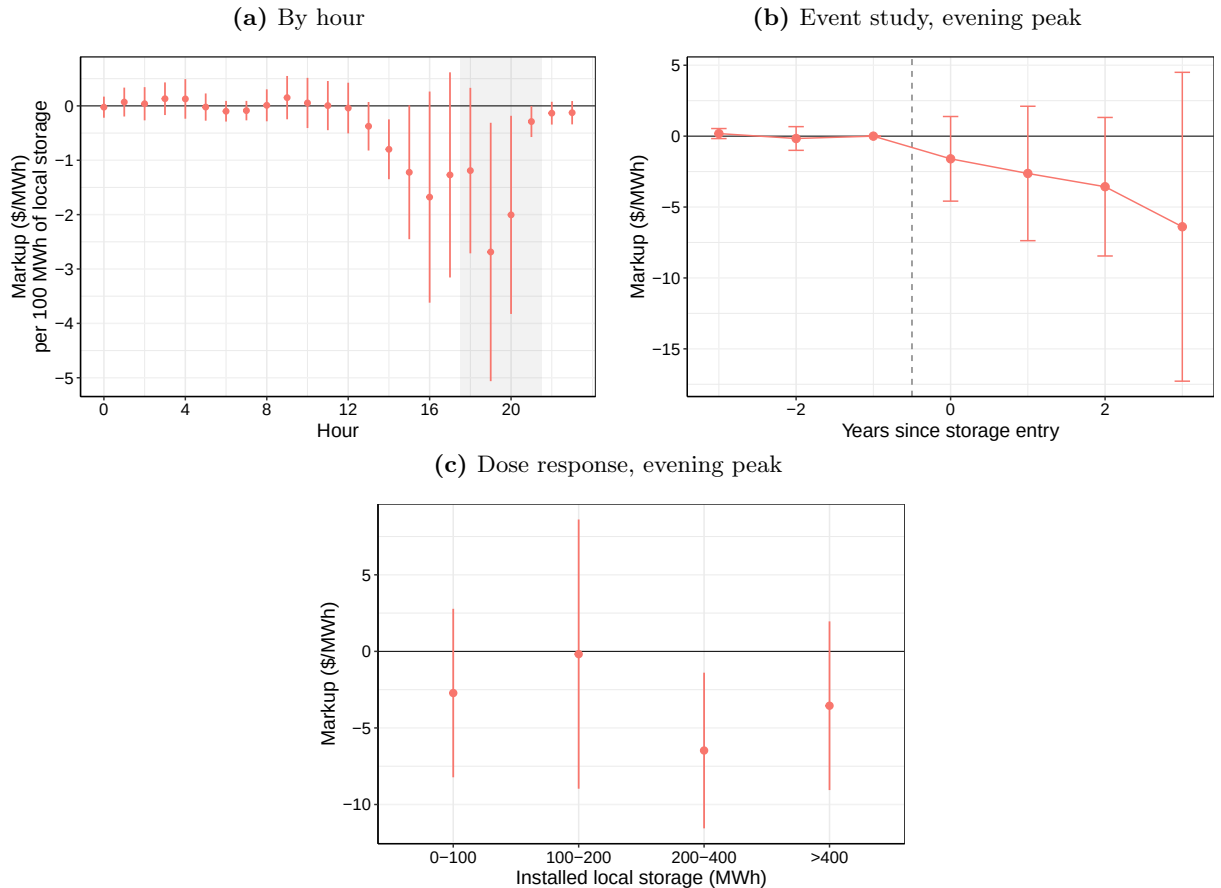
estimates confirm that generator output is responsive to marginal costs, consistent with standard optimal bidding behavior, and that storage disproportionately displaces higher-cost units. The two-stage estimates confirm the displacement and the cost gradient, and they remain robust even when excluding observations near generators' operational limits. Furthermore, estimates of the residual demand slope demonstrate that storage flattens local residual demand, directly confirming the slope channel. For the subsequent welfare analysis, I rely on the technology-specific, capacity-weighted two-stage estimates of the capacity factor. Because this specification holds the generator-hour population fixed and retains observations at zero output or physical limits, it correctly aggregates to the total change in generation (MWh) that determines avoided emissions and the comparison measure of production cost.

6.2 Market Power Results

This section evaluates how local battery storage affects the markups and operating profits of incumbent natural gas generators, decomposing this profit impact into competitive and market power rents. I first present the treatment effects on generator markups, followed by the hourly and daily impacts on operating profits. After decomposing these profit changes, I conclude by estimating the aggregate rent transfer away from natural gas generators. Throughout this section, all treatment effects are scaled per 100 MWh of installed local storage capacity, and all values are reported in 2020 dollars.

Markups Storage lowers the markups of gas generators in the late afternoon and evening. Figure 7 presents the two-stage estimates on the markup. Panel (a) shows that the markup does not change overnight or in the morning. Markup begins to fall at 1 PM, and the effect is largest at 7 PM, when an additional 100 MWh of local storage lowers the markup by \$2.69 per MWh (s.e. 1.18). The effect is also significant at 2 PM, 8 PM, and 9 PM. Over the evening peak from 5 PM to 9 PM, the markup falls by \$1.49 per MWh on average, about 5 percent of the pre-entry evening markup of \$31 per MWh. Panel (b) presents the corresponding event study estimates for the evening peak. The pre-treatment coefficients are near zero and statistically insignificant, supporting the identifying assumption. The post-entry coefficients grow more negative over time, although they are imprecisely estimated. Finally, Panel (c) demonstrates that this markup compression occurs across all dose bins, yielding an average marginal evening effect of -1.30 (s.e. 0.68) per 100 MWh. The pattern is consistent across technologies (Appendix Figure C.11), with evening effects of -0.95 for combined cycle, -1.52 for simple cycle, and -1.72 for steam turbine generators.

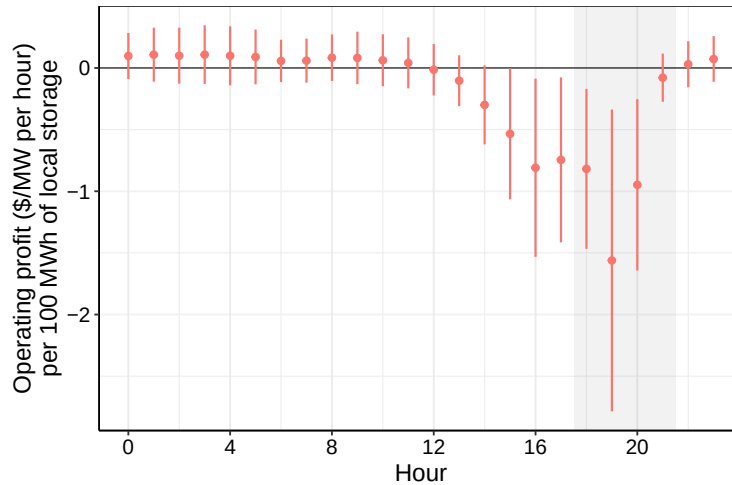
Figure 7: Effect of local storage on markups



Notes. Two-stage difference-in-differences estimates. The markup is the locational marginal price minus marginal cost, set to zero when the generator does not run. Panel (a) reports the effect of 100 MWh of local storage by hour. Panel (b) reports a binary event study for 5 PM to 9 PM, with the year before entry as the reference. Panel (c) reports the effect of each dose bin relative to no storage for 5 PM to 9 PM. The 95% confidence intervals use t_{G-1} critical values and standard errors clustered at the county level.

Operating profit The markup results translate into lost operating profit in the same hours. Figure 8 presents the effect on hourly operating profit per MW of capacity, weighted by capacity. Gas generators lose operating profit from 3 PM to 8 PM, with the largest loss of \$1.56 per MW at 7 PM. Overnight, operating profit remains the same. Summed over the day, the hourly effects imply a loss of about \$4.8 per MW-day.

Figure 8: Effect of local storage on hourly operating profit

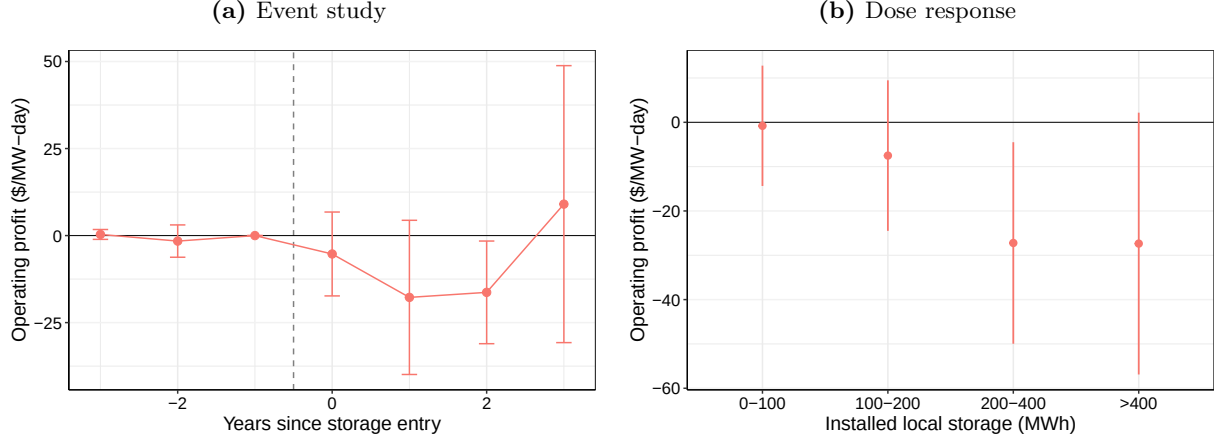


Notes. Two-stage difference-in-differences estimates of the effect of 100 MWh of local storage on hourly operating profit per MW of nameplate capacity, weighted by capacity. The 95% confidence intervals use t_{G-1} critical values and county clustering.

The daily estimates confirm this loss. I aggregate the panel to the generator-day and estimate the effect on daily operating profit per MW, weighted by capacity. An additional 100 MWh of local storage lowers daily operating profit by \$5.33 per MW-day (s.e. 1.86), or about 6 percent of the pre-entry average of \$84 per MW-day. This is close to the sum of the hourly effects. Figure 9 presents the event study and the dose response. The coefficients before entry are small and insignificant. After entry, operating profit falls by \$18 per MW-day in the first year and \$16 per MW-day in the second year, and the second-year effect is significant. The dose response shows that the loss grows with the installed dose. Counties with 200 to 400 MWh of storage lose \$27 per MW-day relative to counties without storage, and counties with more than 400 MWh lose the same amount.

Why distinguish competitive rent and market power rent A fall in gas operating profit does not necessarily mean that storage reduces market power. A gas generator earns operating profit for two reasons. It earns a competitive rent when a more expensive unit sets the price, which it would earn even in a competitive market. Storage can lower competitive rent if its discharge lowers the load gas must serve, so a cheaper unit becomes marginal. It earns a market power rent when the price exceeds the competitive level. Storage can lower market power rents if it also flattens residual demand, which narrows the gap between the price and the competitive level. Only the second is a reduction in market power. Separating the two shows whether storage disciplines

Figure 9: Effect of local storage on daily operating profit



Notes. Two-stage difference-in-differences estimates on the generator-day panel. The outcome is daily operating profit per MW of nameplate capacity, weighted by capacity, with generator and load-zone-by-date fixed effects. Panel (a) reports a binary event study with the year before entry as the reference. Panel (b) reports the effect of each dose bin relative to no storage. The 95% confidence intervals use t_{G-1} critical values and standard errors clustered at the county level.

the conduct of gas generators or simply adds supply.

I decompose the two components using the observed marginal costs of gas units following the approach of [Borenstein, Bushnell and Wolak \(2002\)](#). For each local market and 15-minute interval, I stack the available capacity of gas generators from lowest to highest marginal cost and fill the realized gas output from the bottom of the stack. The competitive price p^c is the marginal cost of the last generator needed to serve that output which represents a cost-based dispatch that same gas output would produce, holding the output of other generators fixed. Each generator's operating profit then splits into two parts,

$$(16) \quad \pi_{iht} = \underbrace{(p_{n(i)th}^c - c_{it}) q_{iht}}_{\text{competitive rent}} + \underbrace{(p_{iht} - p_{n(i)th}^c) q_{iht}}_{\text{market power rent}}.$$

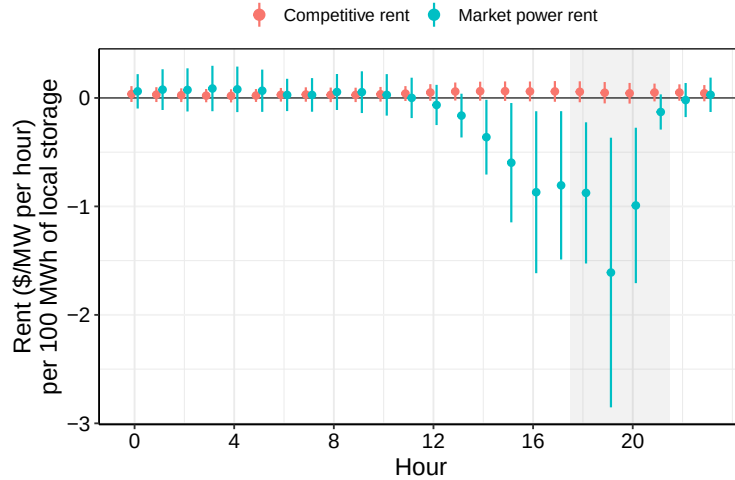
I estimate the effect of storage on each component. Because the estimator is linear, the two effects sum to the effect on operating profit.

This central benchmark likely overstates the market power rent, because measured marginal cost does not factor in startup, maintenance, and ramping costs, which understates the competitive price. I therefore construct a lower bound from the generator that actually sets the price. ERCOT identifies the marginal generator who sets the price in each interval. When a gas generator sets the local price, I use its marginal cost as the competitive price. Because withholding pushes more expensive generators to the margin, the cost of the marginal generator is at or above the true competitive price, so this benchmark understates the market power rent. When no gas generator sets the price, I keep the central benchmark, so the sample composition stays the same. A gas generator sets the local price in 58 percent of intervals.

Rent decomposition The loss in operating profit falls mainly on the market power rent. Figure 10 presents the hourly effects on the two rents under the central benchmark. The market power rent falls from 2 PM to 8 PM, by up to \$1.61 per MW at 7 PM, and accounts for nearly all of the hourly profit loss. The competitive rent changes little in any hour. Table 6 presents the daily effects. Under the central benchmark, the market power rent falls by \$6.19 per MW-day (s.e. 1.74), while the competitive rent rises by an insignificant \$0.86. Under the price-setter benchmark, the market power rent falls by \$3.80 per MW-day (s.e. 1.83), and the competitive rent falls by an insignificant \$1.53. The market power rent therefore accounts for between 71 percent and all of the loss in operating profit, and it is significant under both benchmarks. The estimates by technology tell the same story. Under the price-setter benchmark, the market power rent accounts for 70 percent of the loss for combined cycle generators and 88 percent for simple cycle generators (Appendix Table C.5).

Storage lowers markups without displacing gas from being the price-setter. The probability that a gas generator sets the local price does not change significantly with local storage in any hour (Appendix Figure C.16). If storage reduced gas profits only by adding supply, it would push cheaper units or storage itself to the margin. I find no evidence of this in the data and gas generators continue to set the price, but at a smaller margin above cost. This pattern supports the interpretation that storage disciplines the market power of gas generators.

Figure 10: Effect of local storage on competitive and market power rents by hour



Notes. Two-stage difference-in-differences estimates of the effect of 100 MWh of local storage on the competitive rent and the market power rent in Equation (16) under the central benchmark, per MW of nameplate capacity per hour, weighted by capacity. The two effects sum to the effect on hourly operating profit. The 95% confidence intervals use t_{G-1} critical values and standard errors clustered at the county level.

Aggregate transfer Scaling the daily effects by installed storage and generator capacity gives the aggregate transfer from gas generators. Local storage lowered the operating profit of gas generators by \$46.8 million per year (s.e. 16.3), or about \$4.0 million per year per 100 MWh of storage. Of this,

Table 6: Effect of local storage on daily operating profit and rents

	Operating profit	Competitive rent	Market power rent	Competitive rent (price-setter)	Market power rent (price-setter)
	(1)	(2)	(3)	(4)	(5)
\$/MW-day per 100 MWh	-5.33*** (1.86)	0.86 (0.78)	-6.19*** (1.74)	-1.53 (1.24)	-3.80** (1.83)
Share of profit effect	1.00	-0.16	1.16	0.29	0.71
Annual total (\$ million)	-46.8 (16.3)	7.5 (6.9)	-54.3 (15.3)	-13.4 (10.9)	-33.3 (16.1)
Per 100 MWh of storage (\$ thousand/year)	-3,950 (1,374)	634 (581)	-4,585 (1,291)	-1,134 (918)	-2,816 (1,357)
Observations (generator-days)	677,117				
Counties	50				

Notes. Two-stage difference-in-differences estimates on the generator-day panel, weighted by capacity, with generator and load-zone-by-date fixed effects and the local renewable and temperature controls. Columns (2) and (3) use the central competitive benchmark. Columns (4) and (5) use the price-setter benchmark, which sets the competitive price to the marginal cost of the gas generator setting the local price and keeps the central benchmark when no gas generator sets it. The annual total multiplies the daily effect by installed storage and capacity on treated generator-days and divides by the 6.9 sample years. Dollar amounts are in 2020 dollars, deflated by year with the CPI-U before estimation. The last rows divide the annual total by the average installed storage in gas counties, 1,184 MWh. Standard errors clustered at the county level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

the market power rent accounts for between \$33.3 million per year (s.e. 16.1) under the price-setter benchmark and \$54.3 million per year (s.e. 15.3) under the central benchmark. Aggregating over the technology-specific estimates yields a similar total loss of \$54.4 million, so the aggregate number is robust to pooling across technologies. These amounts are transfers. The rents gas generators lose accrue to consumers through lower prices and to storage owners through arbitrage revenue, and I account for them in the welfare analysis of Section 7.

Summary of market power results Local storage reduces the markups and operating profits of gas generators in the late afternoon and evening, when storage discharges and gas generators would otherwise exercise market power. The daily profit loss of \$5.33 per MW-day per 100 MWh of storage grows with the installed dose. Between 71 percent and all of this loss falls on the market power rent, and gas generators continue to set the price after storage enters. Storage therefore disciplines the market power of gas generators rather than only adding supply.

7 Welfare and Distributional Effects

This section quantifies whether local storage raises total surplus and who gains and loses. I first separate changes in total surplus from transfers among market participants. I then measure each component for the storage installed in counties with gas generation used in my empirical analysis in Sections 6.1 and 6.2. All amounts are annual averages over the sample, in 2020 dollars, the dollar year of the AP4 damages and the social cost of carbon. I deflate prices, costs, and profits by year

with the annual average CPI-U ([U.S. Bureau of Labor Statistics, 2025](#)) before estimation, so the estimated effects are also in 2020 dollars.

7.1 Welfare Accounting

Load is inelastic in the model, so storage changes total surplus only by changing the cost of serving load and the damages from emissions. I write the change in total surplus as

$$(17) \quad \Delta W = V^S + \Delta D^{\text{local}} + \Delta D^{\text{CO}_2} - K,$$

where V^S is the value of the energy storage moves across hours, ΔD are avoided damages, and K is the annualized capital cost of storage.

Value of energy shifting I value storage operations at marginal cost rather than at market prices. Let S_{mth} be the net discharge of storage in county m at hour h of date t , positive when storage discharges. I compute

$$(18) \quad V^S = \sum_{m,t,h} p_{mth}^c S_{mth},$$

where p_{mth}^c is the competitive price of Section 6.2, the marginal cost of the last gas generator in a cost-based dispatch of realized gas output. Discharge substitutes for generation at the marginal cost of the discharge hour, and charging requires generation at the marginal cost of the charging hour. V^S therefore measures the avoided production cost, net of round-trip losses. I use observed storage operations in the data rather than the estimated displacement.

I prefer this approach to using the local displacement estimate. Local storage displaced about 870 GWh of gas generation per year, but storage discharged only 196 GWh. With inelastic load, other generators must supply the difference. Some of this may be gas generation that moves to counties without storage, which the county-level design cannot separate from a true reduction. As a comparison, I also report the avoided production cost implied by the local displacement, with the replacement generation valued at the competitive price.

Avoided damages I compute avoided emissions by multiplying the displaced generation of each generator-hour by that generator’s weekly emission rates of NO_x , SO_2 , and CO_2 from CEMS, and a $\text{PM}_{2.5}$ factor from [U.S. Environmental Protection Agency \(2020\)](#). I value local pollutants with the AP4 source-receptor matrices, which give the damage in each receptor county per ton emitted at each source ([Tschofen, Azevedo and Muller, 2019](#)). I match each generator to its AP4 source by plant code. Fifteen plants, holding 5 percent of gas capacity, have no AP4 source and take the source of the nearest plant of the same technology, in the same county when possible. I value CO_2 at a social cost of carbon of \$190 per metric ton ([EPA, 2023](#)). Both damage measures use the local displacement, so they share its caveat.

Storage cost The investment cost comprises a power cost of 1,440 \$/kW and an energy cost of 461.08 \$/kWh (EIA, 2024). I apply these unit costs to the average installed capacity of storage in counties with gas generation over the sample, 756 MW and 1,157 MWh, an average duration of 1.53 hours. This is the storage whose benefits I estimate, so costs and benefits refer to the same capacity. Applying a capital recovery factor for a 15-year asset life, following the NREL Annual Technology Baseline assumption for utility-scale lithium-ion storage, at a 7% real discount rate gives an annualized cost of about \$178 million per year.¹⁶

Transfers Storage also moves surplus among market participants. Gas producers lose the operating profit estimated in Section 6.2. Storage owners earn arbitrage revenue, $\pi^S = \sum_{m,t,h} \lambda_{mth} S_{mth}$, where λ_{mth} is the price at the storage node. Consumers gain from lower prices. I estimate the effect of storage on the price at gas nodes with the two-stage design, $\hat{\beta}_h^\lambda$ by hour presented in Appendix Figure C.17, and apply it to local load,

$$(19) \quad \Delta CS = - \sum_h \hat{\beta}_h^\lambda \sum_{m,t} B_{mt} L_{mth},$$

where L_{mth} is the load of county m , which I approximate by allocating ERCOT load to counties by population. The remaining generators absorb the rest. Because transfers sum to the change in production cost, I compute their change as the residual, $\Delta\pi^{\text{other}} = V^S - \Delta CS - \Delta\pi^{\text{gas}} - \pi^S$.

7.2 Welfare Results

Table 7 presents the welfare accounting. The first panel reports the change in total surplus, the second the transfers, the third a comparison measure of production cost, and the fourth storage revenue and cost recovery.

Total surplus Storage creates little value by shifting energy at marginal cost. When valued at the competitive price, storage operations slightly raise production costs by \$0.5 million per year. As Figure 1 demonstrates, storage charges overnight and in the morning, when the competitive price is low, and discharges in the evening peak, when it is high. However, the competitive price spread between these periods is small, and storage loses about 19 percent of its energy to round-trip efficiency losses. These two effects roughly cancel each other out. A comparison using the local displacement estimates yields a slightly higher gain of \$5.9 million per year (s.e. 2.5), though this remains small.

Avoided damages represent the largest benefit, though these rely heavily on the local displacement estimates. Avoided local damages total \$12.9 million per year (s.e. 5.7), while avoided CO₂ damages reach \$77.7 million per year (s.e. 32.8). The total benefit is therefore \$90.1 million per

¹⁶I annualize using the capital recovery factor $\text{CRF} = r(1+r)^N / [(1+r)^N - 1]$, with asset life $N = 15$ and real discount rate $r = 0.07$, so that $\text{CRF} \approx 0.110$. The annualized cost excludes fixed operating costs.

Table 7: Welfare and distributional effects of local storage

	Annual (2020 \$ million)	Per 100 MWh of storage (2020 \$ million/year)
<i>Change in total surplus</i>		
Value of storage energy shifting	-0.5	-0.05
Avoided local damages	12.9 (5.7)	1.09 (0.48)
Avoided CO ₂ damages	77.7 (32.8)	6.57 (2.77)
Total benefit, local damages only	12.4 (5.7)	1.04 (0.48)
Total benefit, including CO ₂	90.1 (33.3)	7.61 (2.81)
Annualized storage cost	-178.1	-15.05
Net benefit, including CO ₂	-88.0 (33.3)	-7.44 (2.81)
<i>Transfers among market participants</i>		
Gas producers	-46.8 (16.3)	-3.95 (1.37)
Market power rent, central benchmark	-54.3 (15.3)	-4.58 (1.29)
Market power rent, price-setter benchmark	-33.3 (16.1)	-2.82 (1.36)
Storage owners	5.8	0.49
Consumers	49.9 (49.5)	4.22 (4.18)
Other generators (residual)	-9.5	-0.80
<i>Comparison: production cost from local displacement</i>		
Avoided production cost, local displacement	5.9 (2.5)	0.50 (0.21)
Avoided production cost, replacement at LMP	-11.8	-1.00
<i>Storage owner revenue and cost recovery</i>		
Storage revenue, energy arbitrage	5.8	0.49
Storage revenue, ancillary services	35.6	3.00
Storage revenue, total	41.4	3.49
Share of annualized cost recovered	23%	
Net benefit, ancillary services at their price	-52.4	-4.43

Notes. Annual averages over 2018 to 2024 for storage in counties with gas generation, in 2020 dollars. The value of energy shifting is Equation (18). Avoided damages use the local displacement estimate, CEMS emission rates, and the AP4 source-receptor matrices, with CO₂ at \$190 per metric ton. The storage cost annualizes the EIA investment cost at a 7 percent real discount rate over 15 years. Gas producer transfers are from Table 6. Storage owner transfers cover energy arbitrage only. Consumer surplus is Equation (19). The other generators' change is the residual of the surplus identity. The third panel reports the avoided production cost implied by the local displacement, with replacement generation valued at the competitive price or at the locational marginal price. The fourth panel adds ancillary service revenue from the day-ahead and supplemental markets for the same storage and hours; it does not enter the surplus identity. Per 100 MWh values divide by the average installed storage in gas counties over the estimation sample, 1,184 MWh. Standard errors in parentheses reflect estimation error in the displacement and price effects; observed storage operations and the cost assumptions carry none.

year, or \$12.4 million if CO₂ is excluded. Both estimates assume that the displaced gas generation is not offset by other fossil generation elsewhere. I therefore treat them as upper bounds.

The measured benefits do not cover the capital cost of storage. The annualized capital cost is \$178 million per year, equivalent to \$15.0 million per 100 MWh of storage capacity. Even with avoided damages set at their upper bound, the net benefit is −\$88 million per year. However, this comparison omits the primary service ERCOT storage provides. Storage in counties with gas generation cycled only 0.46 times per day and earned \$35.6 million per year from ancillary services, representing 86 percent of its \$41.4 million total revenue. Ancillary services hold capacity in reserve for grid reliability. Because of this, their social value is absent from standard energy-market accounts. If I value these services at their market clearing prices, which likely overstates their true value when reserve prices exceed the marginal cost of the next available provider, the net benefit rises to −\$52 million per year. These estimates therefore imply that energy arbitrage and its associated local market power effects alone do not justify the investment, not that the investment lowers welfare.

Transfers Storage transfers surplus from gas producers to consumers. Gas producers lose \$46.8 million per year (s.e. 16.3), of which the market power rent accounts for \$33.3 million to \$54.3 million. Consumers gain \$49.9 million per year from lower prices, close to the gas producers' loss. The price effects are negative in every hour but individually imprecise, so the consumer gain is not statistically significant (s.e. 49.5). Storage owners earn \$5.8 million per year from energy arbitrage, or \$8 per kW-year, and the remaining generators lose \$9.5 million as the residual.

Storage owners earn their revenue from the market power of gas generators. Valued at the competitive price, the same operations that earn \$5.8 million are worth −\$0.5 million. The difference is the markup of the marginal gas generators. As storage erodes those markups, it also erodes the price spread. This suggests that the private return to storage arbitrage declines as storage entry disciplines gas generators.

Private cost recovery Ancillary services, not energy arbitrage, provide most of storage revenue. Together with arbitrage, they cover about 23 percent of the annualized capital cost of the storage in counties with gas generation. The gap is partly a feature of the sample, which excludes Winter Storms Uri and Elliott, when reserve and energy prices reached their caps. Storage in these counties earned \$82 million from ancillary services in these 12 days, about 22 percent of all its revenue over the sample period.

Distribution of avoided damages The avoided local damages accrue mostly outside the counties that host storage. Storage counties receive 29 percent of the avoided damages, the rest of Texas 18 percent, and other states 52 percent, because fine particles formed from Texas emissions travel downwind. Per resident, the benefit is heavily concentrated in Texas: \$0.23 per person per year in counties with storage and \$0.20 in the rest of Texas, compared to just \$0.02 outside the state.

Appendix Figure C.18 maps these avoided damages across Texas counties, both in total and per capita.

Summary of welfare results Local storage entry in ERCOT has primarily served to redistribute surplus. It transferred roughly \$47 million per year away from incumbent gas generators, most of it market power rent, and consumers captured the majority of this transfer. The value of storage derived from shifting energy at marginal cost is near zero, and its total measured benefits, even when evaluating avoided damages at their upper bound, fall substantially short of its annualized capital cost. The competitive effect of storage is therefore fundamentally distributional rather than a driver of efficiency gains in the wholesale energy market. Furthermore, the private return to energy arbitrage relies on the very market power that storage capacity inherently disciplines.

8 Conclusion

This paper shows that battery storage disciplines the market power of natural gas generators. Using the staggered rollout of county-level storage capacity in ERCOT, I find that local storage makes the residual demand of nearby gas generators more elastic and displaces their output in the afternoon and evening. It lowers their evening markups and their daily operating profits by about 6 percent per 100 MWh of storage. Between 71 percent and all of the profit loss is market power rent, and gas generators continue to set the price at a smaller margin. These results support a simple mechanism: storage flattens the residual demand of a strategic generator and raises the cost of withholding output.

The competitive effect of storage moved about \$47 million per year from gas producers, most of it market power rent, to consumers. By contrast, storage created little value by shifting energy at marginal cost. Its arbitrage revenue comes from the markups it erodes, and it earns most of its revenue from ancillary services rather than energy. Even with ancillary services, storage revenue covers about a quarter of its annualized capital cost over the sample.

These results carry three policy implications. First, evaluations of storage should count its competitive effect. Standard cost-benefit analyses value storage by the production cost and emissions it avoids. In my setting, both are small relative to the transfer from gas producers to consumers, so omitting competition understates what consumers gain from storage. Second, storage complements market power mitigation. ERCOT combines permissive offer caps with energy-only pricing, and storage lowers markups in the congested local markets where offer mitigation is hardest to target. Interconnection and siting policies that speed storage into these markets would deliver competitive benefits beyond the energy it supplies. Third, consistent with results from prior works (Karaduman, 2023; Butters, Dorsey and Gowrisankaran, 2025), energy arbitrage alone cannot justify storage investment, because its revenue shrinks as storage erodes markups. The design and pricing of ancillary service and reliability products will therefore shape how much storage enters and, through entry, how much competitive discipline it provides.

My analysis has limitations that point to future work. First, the county-level design measures local effects. If gas generation shifts to counties without storage, the local displacement and the avoided emissions overstate the system-wide effects. The welfare accounts also omit the reliability value of ancillary services. Second, I abstract away from storage ownership, and my estimates average across owners. My model implies that an owner of both storage and generation discharges less, because discharging lowers the price its other generation receives. Storage owned by incumbent gas generators or by large renewable portfolios may therefore discipline markups less than storage owned by independent firms. Third, I abstract away from curtailment. I value the energy storage shifts at the marginal cost of gas generation. When wind and solar output exceeds the available transmission capacity, storage instead charges with energy that would otherwise be curtailed and has a marginal cost near zero. In these hours, my estimates understate the value of storage. Curtailment is concentrated in the West and South, where storage is common but gas generators are few, so this value likely falls largely outside my sample. Finally, the estimates reflect storage at its current penetration. As storage grows, it may set prices more often and exercise market power of its own.

References

- Ahmed, Mayesha, Caleb Crain, Shreya Maggo, and Nabila Safitri. 2026. “Replacing California’s Peaker Plants with Long-Duration Energy Storage.”
- Andrés-Cerezo, David, and Natalia Fabra. 2023. “Storing Power: Market Structure Matters.” *The RAND Journal of Economics*, 54(1): 3–53.
- Andres-Cerezo, David, and Natalia Fabra. 2025. “Storage and Renewable Energies: Friends or Foes?” *Working Paper*.
- Arcidiacono, Peter, Paul B. Ellickson, Carl F. Mela, and John D. Singleton. 2020. “The Competitive Effects of Entry: Evidence from Supercenter Expansion.” *American Economic Journal: Applied Economics*, 12(3): 175–206.
- Arkolakis, Costas, and Conor Walsh. 2024. “Clean Growth.” *NBER WORKING PAPER SERIES*.
- Bester, C. Alan, Timothy G. Conley, and Christian B. Hansen. 2011. “Inference with Dependent Data Using Cluster Covariance Estimators.” *Journal of Econometrics*, 165(2): 137–151.
- Borenstein, Severin, James B Bushnell, and Frank A Wolak. 2002. “Measuring Market Inefficiencies in California’s Restructured Wholesale Electricity Market.” *American Economic Review*, 92(5): 1376–1405.
- Borusyak, Kirill, Xavier Jaravel, and Jann Spiess. 2024. “Revisiting Event-Study Designs: Robust and Efficient Estimation.” *The Review of Economic Studies*, 91(6): 3253–3285.
- Bushnell, James, and Kevin Novan. 2021. “Setting with the Sun: The Impacts of Renewable Energy on Conventional Generation.” *Journal of the Association of Environmental and Resource Economists*, 8(4): 759–796.
- Bushnell, James B, Erin T Mansur, and Celeste Saravia. 2008. “Vertical Arrangements, Market Structure, and Competition: An Analysis of Restructured US Electricity Markets.” *American Economic Review*, 98(1): 237–266.
- Busso, Matias, and Sebastian Galiani. 2019. “The Causal Effect of Competition on Prices and Quality: Evidence from a Field Experiment.” *American Economic Journal: Applied Economics*, 11(1): 33–56.
- Butters, R. Andrew, Jackson Dorsey, and Gautam Gowrisankaran. 2025. “Soaking up the Sun: Battery Investment, Renewable Energy, and Market Equilibrium.” *Econometrica*, 93(3): 891–927.
- C3S. 2018. “ERA5 Hourly Data on Single Levels from 1940 to Present.”
- Callaway, Brantly, and Pedro H.C. Sant’Anna. 2021. “Difference-in-Differences with Multiple Time Periods.” *Journal of Econometrics*, 225(2): 200–230.
- Callaway, Brantly, Andrew Goodman-Bacon, and Pedro H C Sant’Anna. 2024. “Difference-in-Differences with a Continuous Treatment.”
- Colin Cameron, A., and Douglas L. Miller. 2015. “A Practitioner’s Guide to Cluster-Robust Inference.” *Journal of Human Resources*, 50(2): 317–372.
- De Chaisemartin, Clément, and Xavier D’Haultfœuille. 2020. “Two-Way Fixed Effects Estimators with Heterogeneous Treatment Effects.” *American Economic Review*, 110(9): 2964–2996.
- Demirer, Mert, and Ömer Karaduman. 2026. “Do Mergers and Acquisitions Improve Efficiency? Evidence from Power Plants.” *Journal of Political Economy*.
- Doshi, Gaurav. 2024. “Wiring America: Electricity Grid Expansion and Fossil Fuel Market Power.”
- Dudek, Matthew. 2026. “Smoothing the Green Transition: Which Battery Locations Provide

- the Greatest Returns to Renewable Generation in California.” *Working Paper*.
- EIA.** 2024. “EIA - U.S. Battery Storage Market Trends.” <https://www.eia.gov/analysis/studies/electricity/batterystorage/>.
- Ellison, Glenn, and Sara Fisher Ellison.** 2011. “Strategic Entry Deterrence and the Behavior of Pharmaceutical Incumbents Prior to Patent Expiration.” *American Economic Journal: Microeconomics*, 3(1): 1–36.
- EPA.** 2023. “EPA Report on the Social Cost of Greenhouse Gases: Estimates Incorporating Recent Scientific Advances.” EPA.
- ERCOT.** 2025. “Fact Sheet.”
- Fabra, Natalia, and Imelda.** 2023. “Market Power and Price Exposure: Learning from Changes in Renewable Energy Regulation.” *American Economic Journal: Economic Policy*, 15(4): 323–358.
- Fabra, Natalia, and Mar Reguant.** 2014. “Pass-Through of Emissions Costs in Electricity Markets.” *American Economic Review*, 104(9): 2872–2899.
- Fell, Harrison, Daniel T. Kaffine, and Kevin Novan.** 2021. “Emissions, Transmission, and the Environmental Value of Renewable Energy.” *American Economic Journal: Economic Policy*, 13(2): 241–272.
- Ganapati, Sharat, Joseph S. Shapiro, and Reed Walker.** 2020. “Energy Cost Pass-Through in US Manufacturing: Estimates and Implications for Carbon Taxes.” *American Economic Journal: Applied Economics*, 12(2): 303–342.
- Gardner, John, Neil Thakral, Linh T Tô, and Luther Yap.** 2024. “Two-Stage Differences in Differences.”
- Goodman-Bacon, Andrew.** 2021. “Difference-in-Differences with Variation in Treatment Timing.” *Journal of Econometrics*, 225(2): 254–277.
- Goolsbee, Austan, and Chad Syverson.** 2008. “How Do Incumbents Respond to the Threat of Entry? Evidence from the Major Airlines*.” *Quarterly Journal of Economics*, 123(4): 1611–1633.
- Hausman, Catherine.** 2025. “Power Flows: Transmission Lines, Allocative Efficiency, and Corporate Profits.” *American Economic Review*.
- Hortaçsu, Ali, and Steven L. Puller.** 2008. “Understanding Strategic Bidding in Multi-unit Auctions: A Case Study of the Texas Electricity Spot Market.” *The RAND Journal of Economics*, 39(1): 86–114.
- Ito, Koichiro, and Mar Reguant.** 2016. “Sequential Markets, Market Power, and Arbitrage.” *American Economic Review*, 106(7): 1921–1957.
- Jha, Akshaya, and Gordon Leslie.** 2025. “Start-Up Costs and Market Power: Lessons from the Renewable Energy Transition.” *American Economic Review*, 115(2): 690–724.
- Joskow, Paul L.** 2022. “From Hierarchies to Markets and Partially Back Again in Electricity: Responding to Decarbonization and Security of Supply Goals.” *Journal of Institutional Economics*, 18(2): 313–329.
- Karaduman, Ömer.** 2023. “Economics of Grid-Scale Energy Storage in Wholesale Electricity Markets.” *Working Paper*.
- Kirkpatrick, A Justin.** 2025. “Estimating the Congestion Benefits of Batteries on Electricity Grids When Network Connections Are Unobserved.” *Working Paper*.
- Lamp, Stefan, and Mario Samano.** 2022. “Large-Scale Battery Storage, Short-Term Market Outcomes, and Arbitrage.” *Energy Economics*, 107: 105786.
- LaRiviere, Jacob, and Xueying Lyu.** 2022. “Transmission Constraints, Intermittent Renewables and Welfare.” *Journal of Environmental Economics and Management*, 112: 102618.
- Mathrani, Amit.** 2025. “Texas: A High Stakes Frontier for US Battery Energy Storage Systems.” <https://www.rabobank.com/knowledge/d011484585-texas-a-high-stakes-frontier->

- for-us-battery-energy-storage-systems.*
- Mercadal, Ignacia.** 2022. “Dynamic Competition and Arbitrage in Electricity Markets: The Role of Financial Players.” *American Economic Journal: Microeconomics*, 14(3): 665–699.
- Miller, David, and Mia Nakajima.** 2024. “Differences in Battery Participation in the CAISO and ERCOT Markets - Gridmatic.” <https://www.gridmatic.com/differences-in-battery-participation-in-the-caiso-and-ercot-markets/>.
- NOAA.** 2024. “Global Surface Hourly (Integrated Surface Database).”
- NREL.** 2024. “National Solar Radiation Database (NSRDB).”
- Reaser, Robert, and Reid B. Taylor.** 2025. “The Short-Run Impacts of Grid-Scale Storage Entry in the Texas Electricity Market.” *Working Paper*.
- Reguant, Mar.** 2014. “Complementary Bidding Mechanisms and Startup Costs in Electricity Markets.” *The Review of Economic Studies*, 81(4): 1708–1742.
- Ryan, Nicholas.** 2021. “The Competitive Effects of Transmission Infrastructure in the Indian Electricity Market.” *American Economic Journal: Microeconomics*, 13(2): 202–242.
- Sun, Liyang, and Sarah Abraham.** 2021. “Estimating Dynamic Treatment Effects in Event Studies with Heterogeneous Treatment Effects.” *Journal of Econometrics*, 225(2): 175–199.
- Taylor, Reid, and Erich Muehlegger.** 2026. “The Effects of Competition in the Retail Gasoline Industry.” *Journal of the Association of Environmental and Resource Economists*.
- Tschofen, Peter, Inês L. Azevedo, and Nicholas Z. Muller.** 2019. “Fine Particulate Matter Damages and Value Added in the US Economy.” *Proceedings of the National Academy of Sciences*, 116(40): 19857–19862.
- U.S. Bureau of Labor Statistics.** 2025. “Historical Consumer Price Index for All Urban Consumers ({CPI-U}): U.S. City Average, All Items.” <https://www.bls.gov/cpi/tables/supplemental-files/historical-cpi-u-202412.pdf>.
- U.S. Environmental Protection Agency.** 2020. “Estimating Particulate Matter Emissions for eGRID.” Clean Air Markets Division Draft White Paper.
- Vermillion, Brandt.** 2024. “Battery Energy Storage in ERCOT’s Real-Time Market: What’s Changing? - Research.” <https://modoenergy.com/research/en/ercot-battery-energy-storage-systems-real-time-prices-offers-owners-eolian-engie-jupiter-power>.
- Vermillion, Brandt.** 2025. “ERCOT: Annual Battery Energy Storage Revenue Report 2024 - Research.” <https://modoenergy.com/research/en/ercot-battery-energy-storage-revenues-energy-markets-ancillary-services-capacity-location-owners-operators-annual-report-2024>.
- Visaggio, Pietro.** 2025. “Pairing Batteries with Renewables: How Ownership Shapes Operational Incentives and Market Outcomes.” *Working Paper*.
- Woerman, Matt.** 2023. “Market Size and Market Power: Evidence from the Texas Electricity Market.” *Working Paper*.
- Yitzhaki, Shlomo.** 1996. “On Using Linear Regressions in Welfare Economics.” *Journal of Business & Economic Statistics*.

Supplementary Appendix

A Detailed Theoretical Derivations

This appendix extends the model of Section 4 in three directions. It shows how transmission congestion determines how much of a battery's discipline reaches a gas generator, generalizes storage from a price taker to a strategic actor, and derives how ownership changes the discipline storage provides. Throughout, I keep the notation of the main text.

A.1 Congestion and the Local Market

Congestion determines how much of a remote battery's discipline reaches the gas firm, which is why I define the market locally. Consider two regions, W and E , joined by a line of capacity $\bar{F}$ that carries flow F from W to E . Region E holds the strategic gas firm f , a fringe $S_E^c(p_E) = \alpha_E + \gamma_E p_E$, and load D_E . Region W holds a fringe $S_W^c(p_W) = \alpha_W + \gamma_W p_W$, load D_W , and a storage fleet with level $\bar{s}_t$ and slope σ_t . Nodal balance and the line constraint are

$$(A.1) \quad (E) \quad q_{ft} + \alpha_E + \gamma_E p_E + F = D_E,$$

$$(A.2) \quad (W) \quad \alpha_W + (\gamma_W + \sigma_t)p_W + \bar{s}_t = D_W + F,$$

$$(A.3) \quad F \leq \bar{F}, \quad p_E - p_W = \lambda \geq 0, \quad \lambda(\bar{F} - F) = 0,$$

so the price gap equals the congestion shadow price λ .

When the line is slack, remote storage reaches the gas firm in full. With $\lambda = 0$, both regions clear at one price p , and adding (A.1) and (A.2) eliminates the flow,

$$(A.4) \quad q_{ft} = (D_E + D_W) - (\alpha_E + \alpha_W + \bar{s}_t) - (\gamma_E + \gamma_W + \sigma_t)p.$$

The firm faces the slope $\gamma_E + \gamma_W + \sigma_t$, because the flow adjusts to carry storage output into region E .

When the line binds, remote storage does not reach the gas firm at all. The import into E is fixed at $F = \bar{F}$, so (A.1) gives $q_{ft} = D_E - \alpha_E - \gamma_E p_E - \bar{F}$, and the firm faces the slope γ_E alone. Storage clears within W , lowering p_W and widening λ , but it leaves the firm's output and markup unchanged. Storage located in E , by contrast, enters (A.1) directly and gives the firm the slope $\gamma_E + \sigma_t$ regardless of the line.

These three cases order storage by its share of the firm's competitive set. The share is $\sigma_t/(\gamma_E + \sigma_t)$ for local storage, the smaller $\sigma_t/(\gamma_E + \gamma_W + \sigma_t)$ for remote storage on a slack line, and zero for remote storage behind a binding constraint. Local storage therefore delivers the full effect, remote storage an attenuated one, and constrained remote storage none. Storage entry can itself change congestion, because storage in W lowers p_W and raises desired exports. I abstract from this by assuming no constraint is marginally binding, following [Woerman \(2023\)](#). The same logic implies that the effect of storage should scale with its size relative to the local market, which motivates

defining treatment at the county level.

A.2 Strategic Storage

Strategic storage weakens the discipline that price-taking storage imposes, and the price-taking model is its limiting case. I follow [Andres-Cerezo and Fabra \(2025\)](#); [Andrés-Cerezo and Fabra \(2023\)](#), in which storage internalizes the price effect of its own dispatch, and focus on the discharge period. Netting out the fringe, the price is $p = (a - q_f - s)/\gamma$ with $a = D_H - \alpha_H$. The gas firm has marginal cost c_f , and storage has marginal cost $m_s = p_L/\eta$.

A conduct parameter $\theta \geq 0$ indexes how much storage withholds. Storage perceives marginal revenue $p - \theta s/\gamma$, so the two first-order conditions are

$$(A.5) \quad p - c_f = \frac{q_f}{\gamma} \quad (\text{gas firm}), \quad p - m_s = \theta \frac{s}{\gamma} \quad (\text{storage}).$$

Storage is a price taker when $\theta = 0$ and a Cournot competitor when $\theta = 1$. Solving,

$$(A.6) \quad s^*(\theta) = \frac{a + \gamma c_f - 2\gamma m_s}{1 + 2\theta}, \quad q_f^*(\theta) = \frac{1}{2}(a - s^* - \gamma c_f), \quad \mu_f^*(\theta) = \frac{q_f^*}{\gamma}.$$

More strategic storage discharges less and leaves a higher gas markup,

$$(A.7) \quad \frac{\partial s^*}{\partial \theta} < 0, \quad \frac{\partial \mu_f^*}{\partial \theta} = \frac{a + \gamma c_f - 2\gamma m_s}{\gamma(1 + 2\theta)^2} > 0 \quad \text{whenever } s^* > 0.$$

Competitive storage delivers the largest markup reduction. As θ grows, storage withholds more, and the gas markup returns toward its level without storage. Storage that is present but strategic can therefore provide little discipline. The main text studies the case $\theta = 0$, in which the slope σ_t comes from aggregating heterogeneous price-taking batteries.

A.3 Ownership

Portfolio ownership reduces the discipline storage provides through two wedges. Suppose the storage owner also sells generation G_H in the discharge period and G_L in the charge period, and its storage is large enough to move prices, $\partial p_t / \partial s_t = -1/E_t$. Maximizing portfolio profit gives the first-order condition

$$(A.8) \quad \underbrace{\eta p_H - p_L}_{\text{arbitrage spread}} + \underbrace{\left(\frac{s_L}{E_L} - \eta \frac{s_H}{E_H} \right)}_{\text{storage-side market power}} + \underbrace{\left(\frac{G_L}{E_L} - \eta \frac{G_H}{E_H} \right)}_{\text{portfolio}} = 0.$$

A price-taking battery with no other generation keeps only the arbitrage spread and discharges fully, as in [Section 4.1](#). The other two terms reduce discharge.

The storage-side term makes a large storage owner withhold. Because $s_L < 0 < s_H$, the term is negative: the owner recognizes that discharging lowers the peak price and charging raises the

off-peak price, so it discharges less and responds less to price.

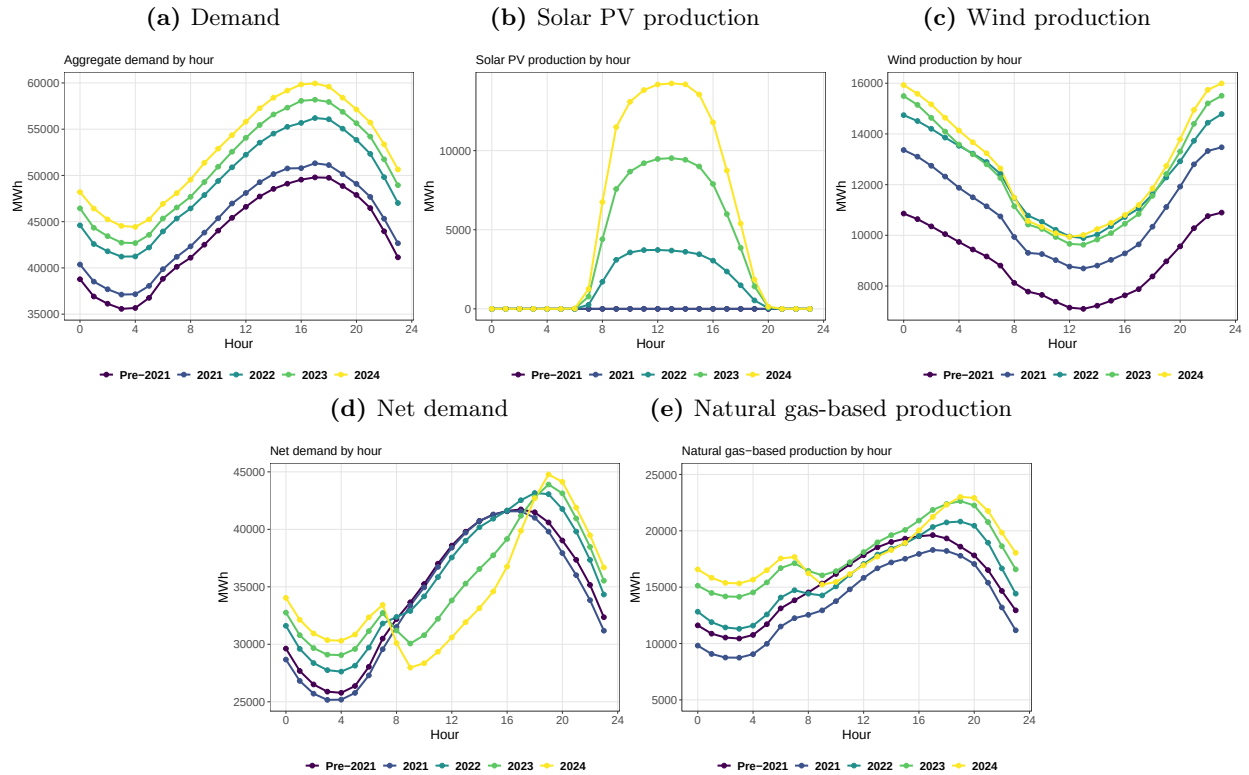
The portfolio term makes an owner with peak generation withhold. Its peak component, $-\eta G_H/E_H$, is negative, because discharging lowers the price at which the owner's other generation sells. This applies to owners of gas generation and equally to owners of wind or solar, whose output also clears at the peak price. A battery co-located with solar faces a related limit, because its charging is tied to the solar profile, which lowers its responsiveness to price.

Both wedges point in the same direction. Storage owned by firms with large generation portfolios or large storage positions should discharge less and impose less discipline on gas markups than storage owned by independent, storage-only firms.

B Additional Descriptive Statistics

B.1 Trends in Demand and Supply Patterns

Figure B.1: Hourly demand and supply patterns in ERCOT



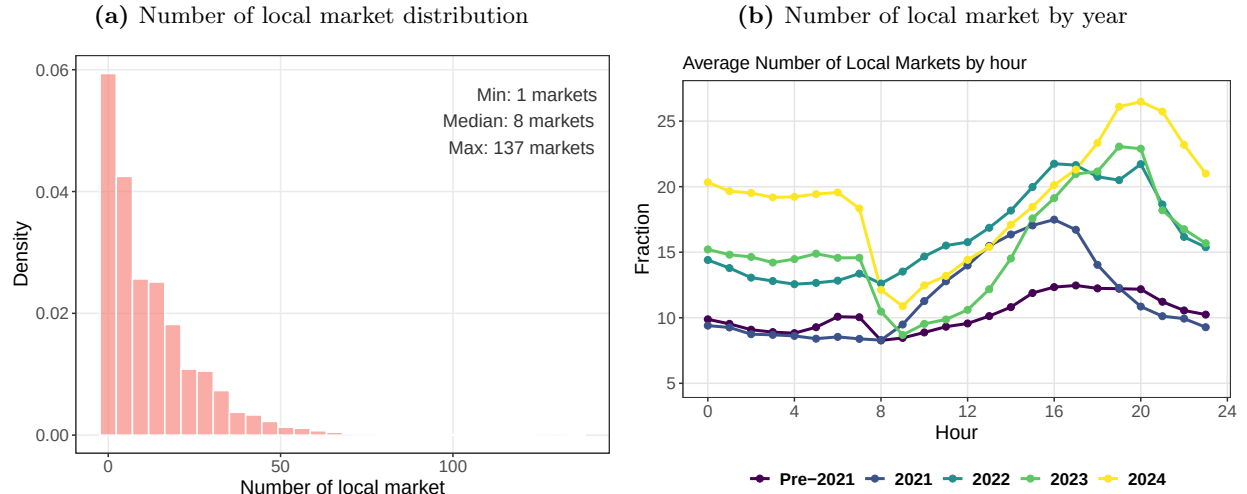
Notes: These figures show the average renewable generation, load, net load, and natural gas-based generation for each hour of the day. The net load is defined as the total load minus the total renewable generation.

Figure B.1 documents the hourly feature of the Texas electricity market over my sample period. First, in Panel B and C, the daily renewable production pattern have changed significantly over the years. Before 2021, the average renewable generation is relatively flat throughout the day, with

its highest production at night due wind domination in the renewable generation. As more solar PV capacity is installed, we see a significant increase in renewable generation during the daytime hours, peaking around 1-2 pm. Figure [B.1a](#) shows that the hourly demand pattern has also evolved over the years, generally increasing across all hour of day. The hourly pattern shows a typical daily cycle with peaks in the morning and evening hours. These two things combined has implications for the net demand pattern in Figure [B.1d](#), which has evolved from a relatively flat pattern to a more pronounced “duck curve” pattern, with a significant dip during the daytime hours when solar PV production is high and a steep ramp in the morning and evening hours when demand is high and solar PV production is low.

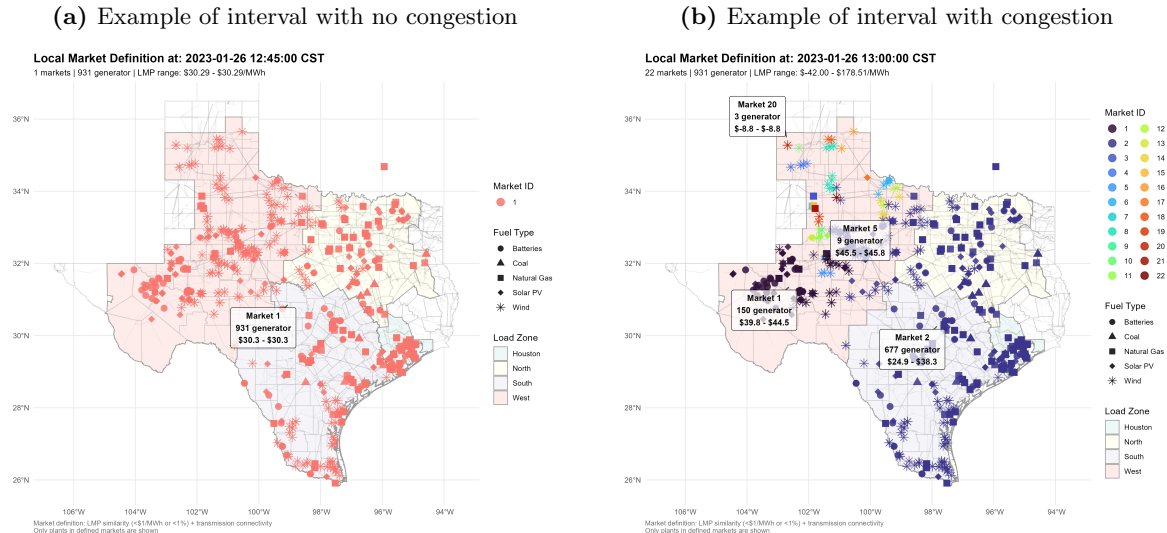
B.2 Local Market Descriptive Statistics

Figure B.2: Local market statistics



Notes: Panel (a) and (b) summarize the distribution of the number of local markets and the trends in hourly variation of the number of local markets by year.

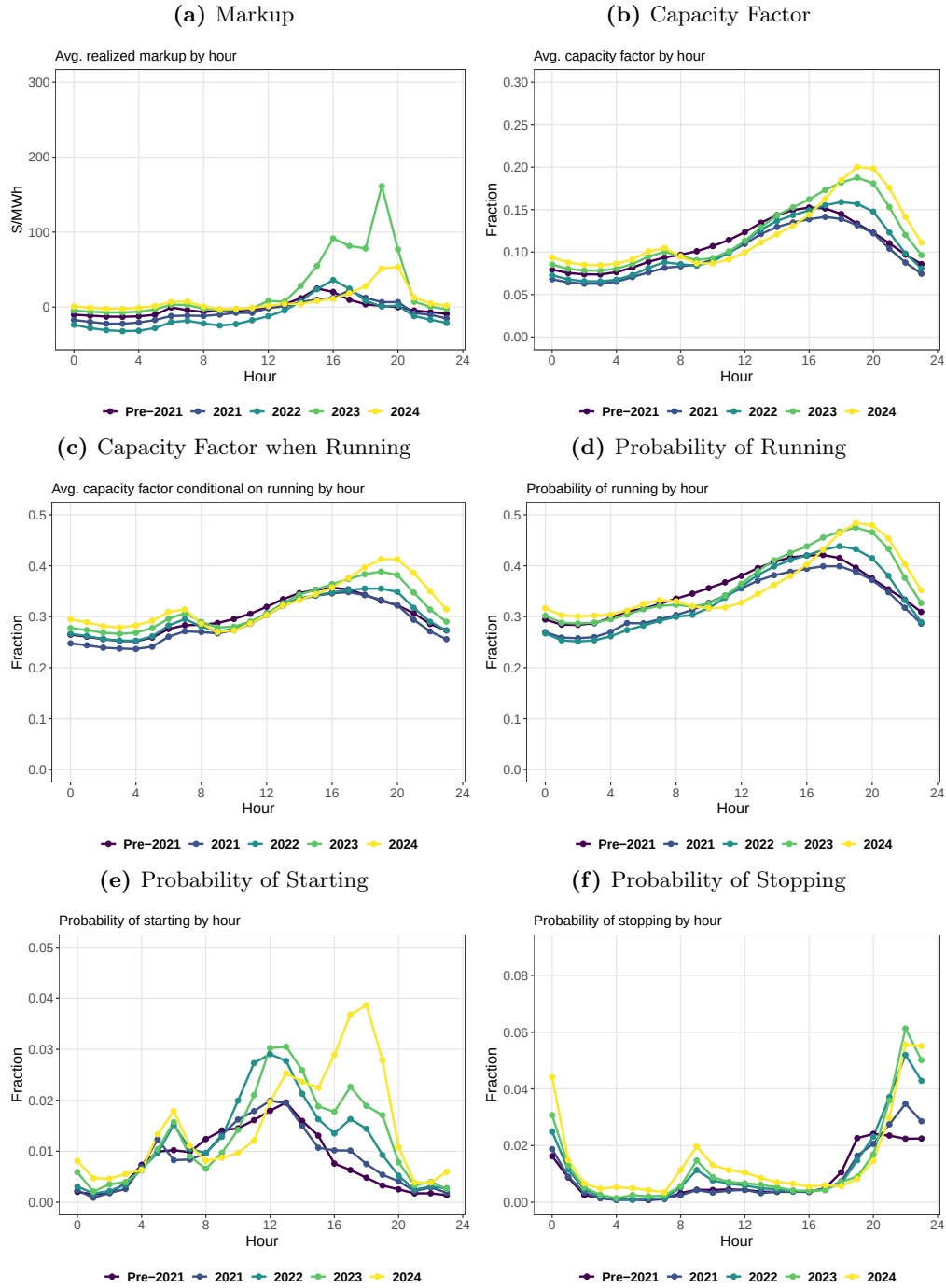
Figure B.3: Example of local market definition



Notes: Panel (a) and (b) show an example of market definition for intervals 15 minutes apart. Each dot is a generator, with colors representing distinct local markets. In panel (c), there is no congestion and all generators are in the same local market, while in panel (d), congestion splits the generators into 22 local markets. Across the sample, the number of local markets varies from 1 to 137, with a median of 8. Figure B.2 presents the distribution of local markets statistics within my sample.

B.3 Natural Gas Generators Descriptive Statistics

Figure B.4: Natural gas unit markup and capacity factor by hour of day



Notes: Each panel shows the average of each variable for each hour of the day, plotted separately for the pre-2021 period and each year from 2021 through 2024. The data exclude period of extreme event.

C Additional Results

C.1 Additional Displacement Results

This appendix reports the full structural estimates, the robustness of the two-stage estimates, and additional results on heterogeneity and the residual demand slope.

Structural estimates Table C.1 reports the structural estimates of Equation (10) across samples, weights, and fixed effects. The IV estimate is -0.79 on interior hours (Column 2) and -0.81 when I include hours at the operating limits (Column 3). Excluding hours within 2 percent of capacity of either limit shrinks it to -0.26 and makes it insignificant (Column 4). Capacity weights give -0.66 (Column 5), and load-zone-by-date-by-hour fixed effects, which absorb the common natural gas price shock, give -0.82 (Column 6). The cost coefficient and the interaction term are negative and significant in every column except the 2 percent band.

Table C.1: Structural estimates across samples, weights, and fixed effects

	OLS	Capacity factor (pp)				
	(1)	(2)	(3)	IV (4)	(5)	(6)
Storage B (100MWh), $\hat{\delta}^{(1)}$	-0.791*** (0.291)	-0.792*** (0.292)	-0.813*** (0.260)	-0.260 (0.175)	-0.657** (0.262)	-0.816** (0.343)
Cost $\tilde{c}$ (centered, \$/MWh), $\hat{\delta}^{(2)}$	-0.067*** (0.017)	-0.068*** (0.016)	-0.066*** (0.018)	-0.081*** (0.012)	-0.099*** (0.017)	-0.069*** (0.019)
Cost $\tilde{c} \times B$, $\hat{\delta}^{(3)}$	-0.016*** (0.003)	-0.016*** (0.003)	-0.019*** (0.003)	0.000 (0.008)	-0.011*** (0.004)	-0.017*** (0.004)
Generator FE	✓	✓	✓	✓	✓	✓
Load zone-Month-of-sample-Hour FE	✓	✓	✓	✓	✓	
Load zone-Date-Hour FE						✓
Local renewable capacity and potential	✓	✓	✓	✓	✓	✓
Local temperature	✓	✓	✓	✓	✓	✓
Weights	None	None	None	None	Capacity	None
Sample	Interior	Interior	Running	Interior 2% band	Interior	Interior
Observations	3,808,488	3,808,488	5,507,103	1,751,707	3,808,488	3,798,131
First-stage F , $\tilde{c}$		332.0	629.0	209.1	411.5	313.7
First-stage F , $\tilde{c} \times B$		2,429.5	2,325.5	1,781.6	530.1	8,829.2
Partial R^2 , $\tilde{c}$		0.385	0.363	0.372	0.408	0.543
Partial R^2 , $\tilde{c} \times B$		0.810	0.802	0.846	0.852	0.835

Notes. Estimates of Equation (10). The outcome is the hourly capacity factor in percentage points. Marginal cost is centered within technology and instrumented by the pre-entry heat rate times the Henry Hub price. Column (3) uses all hours with positive output, and Column (4) drops interior hours within 2 percent of capacity of either operating limit. Standard errors clustered by county in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C.2 reports the structural estimates by technology. None of the technology-specific storage coefficients is statistically significant. The interaction term is negative and significant for combined cycle and simple cycle generators, so storage displaces the higher-cost units within these technologies more. It is positive for steam turbines. Figure C.5 presents the hourly structural estimates. The displacement at mean cost is between 0.2 and 0.4 percentage points for combined cycle generators

and between 0.4 and 1.2 percentage points for simple cycle generators, and it is imprecise in most hours. The cost gradient has the predicted sign in every hour for both technologies. Figure C.6 shows that the instruments explain 30 to 46 percent of the remaining variation in cost and 79 to 87 percent of the remaining variation in the interaction for these two technologies in every hour.

Table C.2: Structural estimates by technology

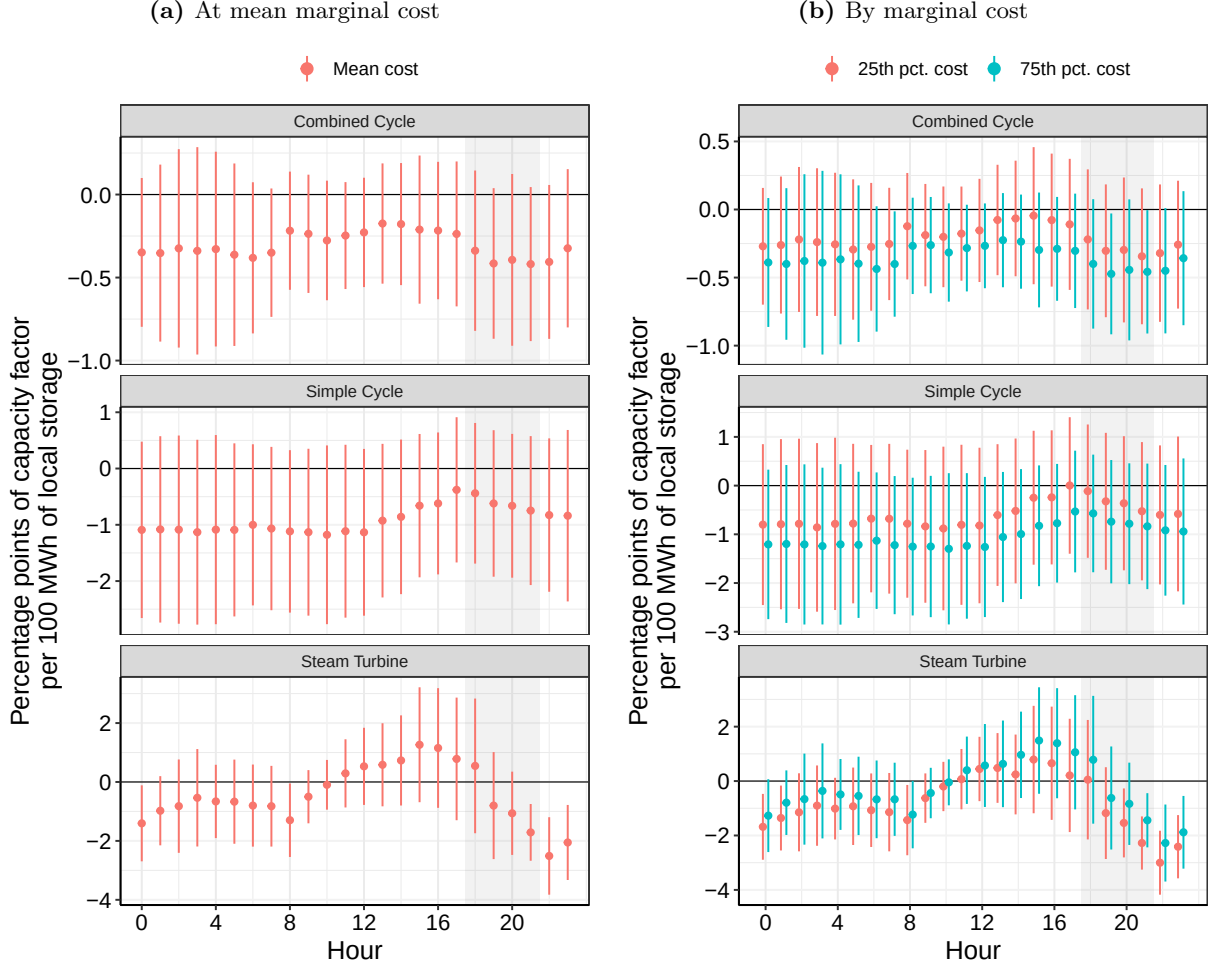
	Capacity factor (pp)			
	All gas (1)	Combined Cycle (2)	Simple Cycle (3)	Steam Turbine (4)
Storage B (100MWh), $\hat{\delta}^{(1)}$	-0.792*** (0.292)	-0.300 (0.208)	-0.958 (0.635)	0.169 (0.547)
Cost $\tilde{c}$ (centered, \$/MWh), $\hat{\delta}^{(2)}$	-0.068*** (0.016)	-0.092*** (0.016)	-0.035 (0.026)	-0.120*** (0.037)
Cost $\tilde{c} \times B$, $\hat{\delta}^{(3)}$	-0.016*** (0.003)	-0.010** (0.004)	-0.024*** (0.004)	0.029*** (0.007)
Generator FE	✓	✓	✓	✓
Load zone-Month-of-sample-Hour FE	✓	✓	✓	✓
Local renewable capacity and potential	✓	✓	✓	✓
Local temperature	✓	✓	✓	✓
Sample	Interior	Interior	Interior	Interior
Observations	3,808,488	1,833,242	1,635,357	339,526
First-stage F , $\tilde{c}$	332.0	254.0	131.0	89.7
First-stage F , $\tilde{c} \times B$	2,429.5	1,822.4	13,639.7	348.2
Partial R^2 , $\tilde{c}$	0.385	0.410	0.347	0.187
Partial R^2 , $\tilde{c} \times B$	0.810	0.843	0.816	0.647

Notes. Estimates of Equation (10) on interior hours, separately by technology. Cost and the instrument are centered within technology. Standard errors clustered by county in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Two-stage estimates across samples Table C.3 reports the pooled two-stage estimates across samples and weights. The storage coefficient is -0.41 on interior hours, -0.33 on hours with positive output, and -0.46 when I exclude hours within 2 percent of either operating limit. It is -0.79 on all hours and -0.91 on all hours with capacity weights. Unlike the IV estimate, the two-stage estimate remains significant when I exclude hours near the operating limits.

Reconciling the structural and two-stage estimates Table C.4 changes one feature at a time by technology and hour group. For combined cycle generators, the four estimates are close in every hour group, so the choice of specification matters little. For simple cycle generators, the two-stage estimate on the interior sample is larger and more precise than the IV estimate, at 0.8 to 1.6 percentage points, and changing the sample to all hours moves the estimate toward the welfare specification. For steam turbines, the estimates on the interior sample are unreliable because storage changes which steam turbines operate in the interior. Table 5 shows that storage moves steam turbines away from maximum output and into the interior, which changes the composition of the interior sample. The estimates on all hours do not have this problem.

Figure C.5: Hourly structural displacement by technology

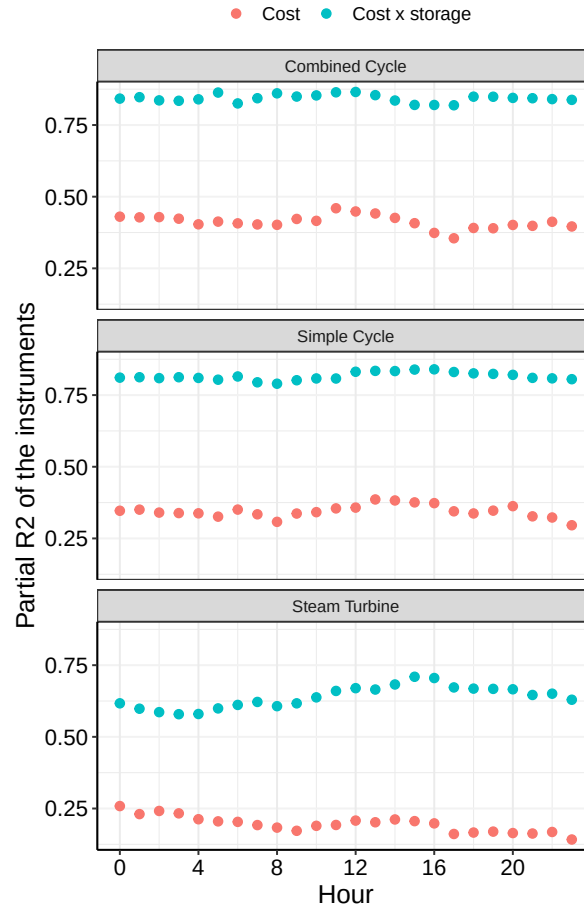


Notes. Estimates of Equation (10) by technology and hour on interior hours, in percentage points of capacity factor per 100 MWh of local storage. Panel (b) evaluates the displacement at the 25th and 75th percentiles of marginal cost within each technology. The 95% confidence intervals use t_{G-1} critical values and with standard errors clustered at the county level.

Local renewable controls Figure C.7 compares the welfare specification with and without the local renewable controls. The controls lower the combined cycle estimates by about 0.1 percentage points in every hour. For simple cycle generators, they remove most of the overnight displacement but leave the afternoon displacement nearly unchanged. Solar capacity that enters alongside storage therefore does not explain the afternoon displacement.

Heterogeneity Figure C.8 presents the displacement for all gas generators by region and by renewable conditions. In the Houston and North load zones, storage lowers the capacity factor by 1.6 percentage points in the evening and 1.2 percentage points at midday, and the evening effect is significant in every hour. In the West and South load zones, the evening effect is 1.1 percentage points and not significant. The displacement is larger on days with above-median solar output, by

Figure C.6: First-stage strength by hour



Notes. Partial R^2 of the excluded instruments for the cost term and for the cost-by-storage interaction in each hourly first stage, by technology.

about 0.4 percentage points at midday and in the evening. It is also larger on days with above-median wind output, especially overnight, when it is 0.8 percentage points against 0.3 percentage points on low-wind days.

Residual demand slope Figure C.9 presents the effect of storage on the measured residual demand slope by technology. The residual demand of combined cycle generators becomes significantly flatter between 6 PM and 11 PM and steeper at 2 PM and 3 PM. The effects for simple cycle and steam turbine generators are imprecise. Figure C.10 presents the event study for the slope by hour group. The coefficients two and three years before entry are small and insignificant in every hour group, so treated and untreated counties followed parallel paths before entry. After entry, the midday slope rises by 14 and 28 MW per \$/MWh in the second and third year. The post-entry effects in the other hour groups are imprecise.

Table C.3: Two-stage estimates across samples and weights

	Capacity factor (pp)				
	2SDD (1)	2SDD (2)	2SDD (3)	2SDD (4)	2SDD (5)
Storage B (100MWh)	-0.407*** (0.134)	-0.330** (0.133)	-0.459** (0.223)	-0.788*** (0.223)	-0.906*** (0.267)
Predetermined cost $\tilde{z} \times B$	-0.016*** (0.004)	-0.009** (0.004)	-0.002 (0.010)	-0.002 (0.004)	-0.004 (0.008)
Generator FE	✓	✓	✓	✓	✓
Load zone-Date-Hour FE	✓	✓	✓	✓	✓
Local renewable capacity and potential	✓	✓	✓	✓	✓
Local temperature	✓	✓	✓	✓	✓
Sample	Interior	Running	Interior 2% band	All hours	All hours
Weights	None	None	None	None	Capacity
Observations	3,690,817	5,398,211	1,665,906	13,575,368	13,575,368
Counties	50	50	50	50	50

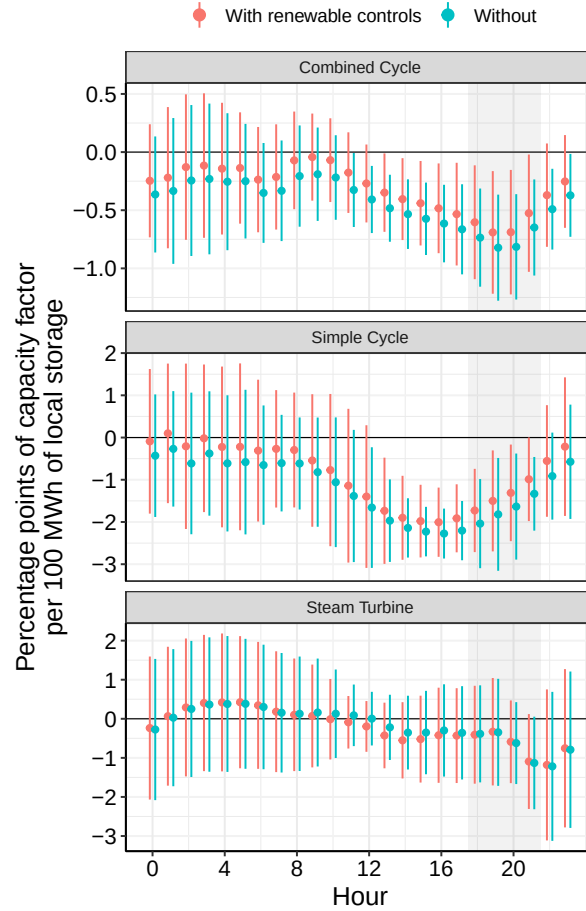
Notes. Two-stage difference-in-differences estimates of the effect of 100 MWh of local storage on the capacity factor, in percentage points, with storage interacted with predetermined cost. Standard errors clustered by county in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C.4: Reconciling the structural and two-stage estimates

	(1) IV Interior Unweighted	(2) 2SDD Interior Unweighted	(3) 2SDD All hours Unweighted	(4) 2SDD All hours Capacity-weighted
<i>Combined Cycle</i>				
Overnight (0–6)	-0.347 (0.261)	-0.260 (0.270)	-0.250 (0.198)	-0.225 (0.257)
Midday (10–16)	-0.220 (0.178)	-0.154 (0.177)	-0.273* (0.146)	-0.288** (0.141)
Evening (17–21)	-0.356 (0.229)	-0.270 (0.244)	-0.483** (0.182)	-0.469** (0.187)
<i>Simple Cycle</i>				
Overnight (0–6)	-1.066 (0.742)	-1.449*** (0.200)	-0.764 (0.670)	-1.195 (0.929)
Midday (10–16)	-0.947 (0.661)	-1.620*** (0.211)	-0.656** (0.310)	-0.770 (0.539)
Evening (17–21)	-0.583 (0.602)	-0.811*** (0.246)	-0.789 (0.590)	-0.797** (0.358)
<i>Steam Turbine</i>				
Overnight (0–6)	-0.847 (0.626)	0.107 (0.174)	-0.468*** (0.064)	-0.444 (0.344)
Midday (10–16)	0.736 (0.652)	3.870** (1.708)	0.268 (0.231)	0.327 (0.675)
Evening (17–21)	-0.282 (0.756)	2.570 (2.252)	-0.649*** (0.199)	0.068 (0.543)

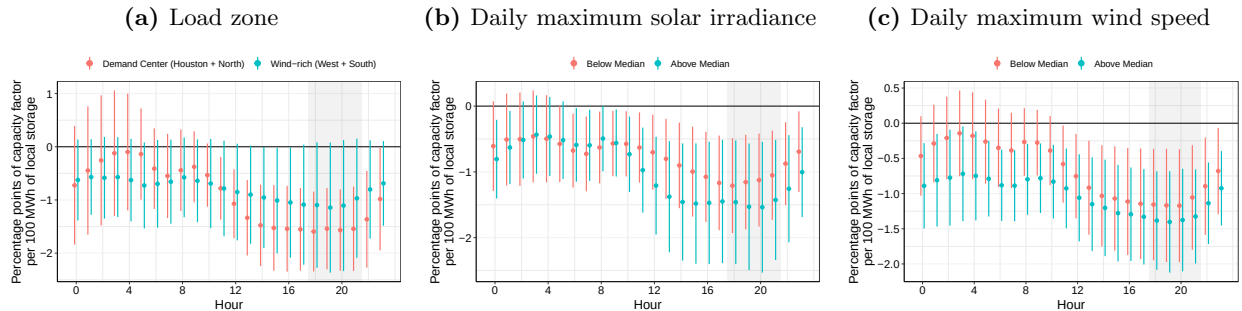
Notes. Effect of 100 MWh of local storage on the capacity factor, in percentage points, by technology and hour group. Each column changes one feature relative to the previous one: the estimator, then the sample, then the weights. Column (4) is the specification used in the welfare analysis. Standard errors clustered by county in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure C.7: Displacement with and without local renewable controls



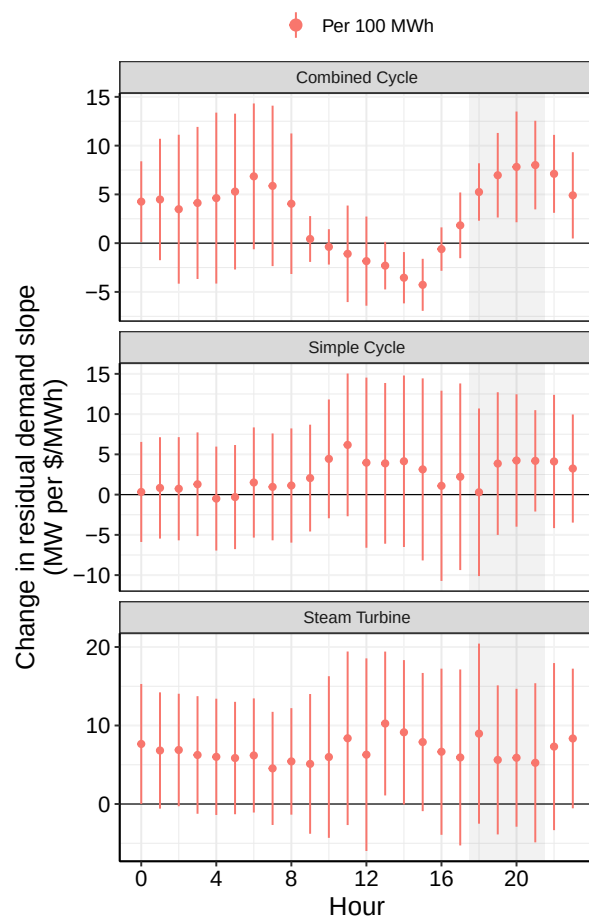
Notes. Two-stage difference-in-differences estimates of the effect of 100 MWh of local storage on the capacity factor, by hour and technology, on all hours and weighted by capacity. The controls are county solar and wind capacity and their potential output, capacity times irradiance or a normalized wind power curve. The 95% confidence intervals use t_{G-1} critical values and with standard errors clustered at the county level.

Figure C.8: Displacement by region and renewable conditions



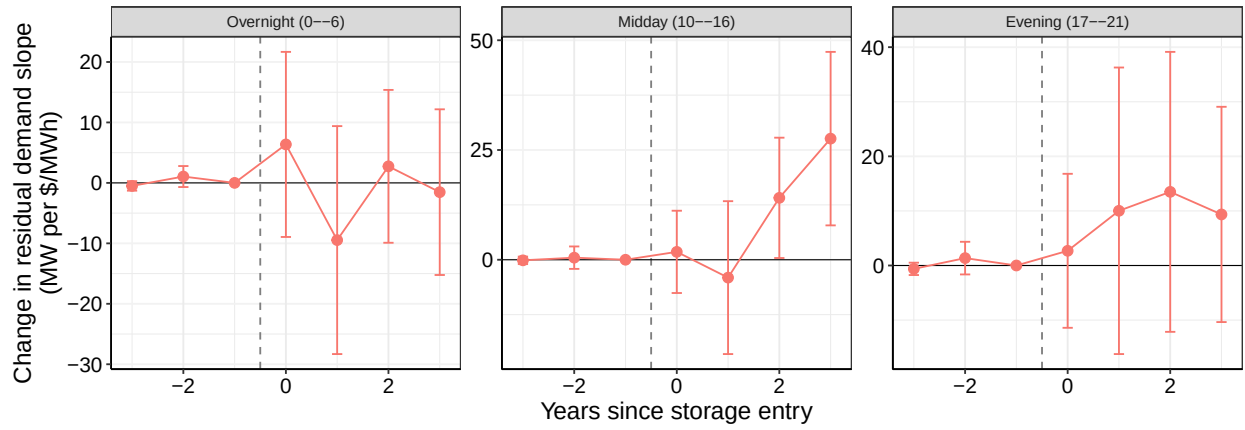
Notes. Two-stage difference-in-differences estimates of the effect of 100 MWh of local storage on the capacity factor of all gas generators, by hour, on all hours and weighted by capacity. Panel (a) splits the sample into the Houston and North load zones and the West and South load zones. Panels (b) and (c) split days at the sample median of statewide daily maximum solar irradiance and wind speed. The 95% confidence intervals use t_{G-1} critical values and with standard errors clustered at the county level.

Figure C.9: Residual demand slope by technology



Notes. Two-stage difference-in-differences estimates of the effect of 100 MWh of local storage on the local residual demand slope faced by each technology, in MW per \$/MWh, by hour. The 95% confidence intervals use t_{G-1} critical values and with standard errors clustered at the county level.

Figure C.10: Event study: residual demand slope



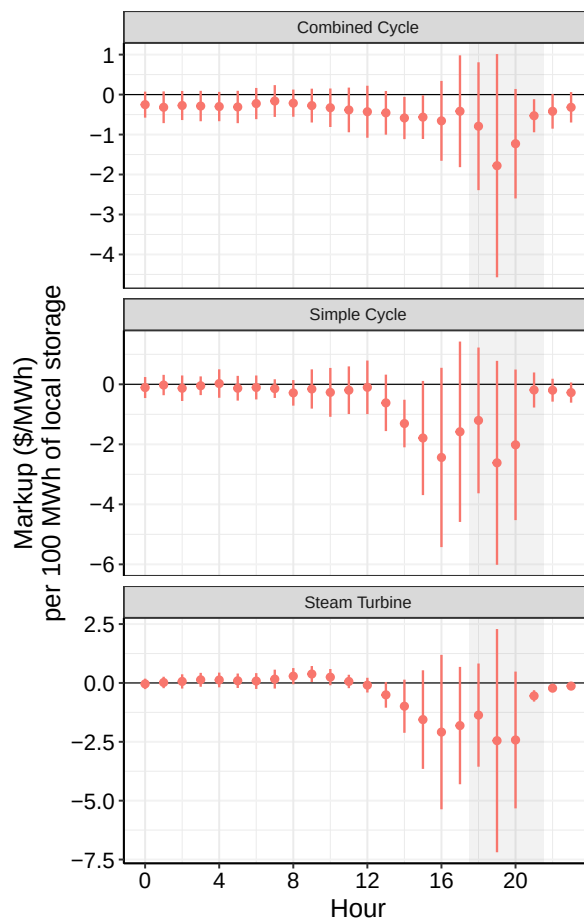
Notes. Binary event study from Equation (14) by hour group, with years relative to the first local storage entry binned at three years. The year before entry is the reference. Intervals use t_{G-1} critical values and with standard errors clustered at the county level.

C.2 Additional Market Power Results

This appendix reports the market power results by technology and the event studies not shown in the main text.

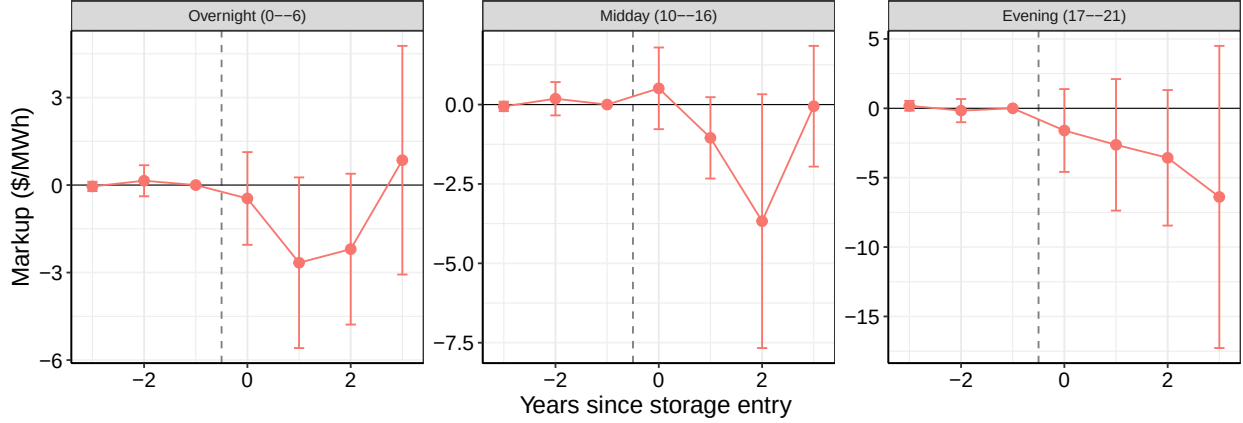
Markups by technology and hour group Figure C.11 presents the hourly markup effects by technology. All three technologies show the same pattern as the pooled estimate, with no change overnight and a fall in the late afternoon and evening. The evening effects average -0.95 for combined cycle, -1.52 for simple cycle, and -1.72 for steam turbine generators. Figure C.12 presents the markup event studies for the overnight, midday, and evening hour groups. The coefficients before entry are small and insignificant in every hour group. After entry, the overnight markup does not change systematically, while the midday and evening markups fall. None of the individual post-entry coefficients is significant.

Figure C.11: Effect of local storage on markups by technology



Notes. Two-stage difference-in-differences estimates of the effect of 100 MWh of local storage on the markup, in \$/MWh, by hour and technology. Intervals use t_{G-1} critical values and with standard errors clustered at the county level.

Figure C.12: Event study: markups by hour group



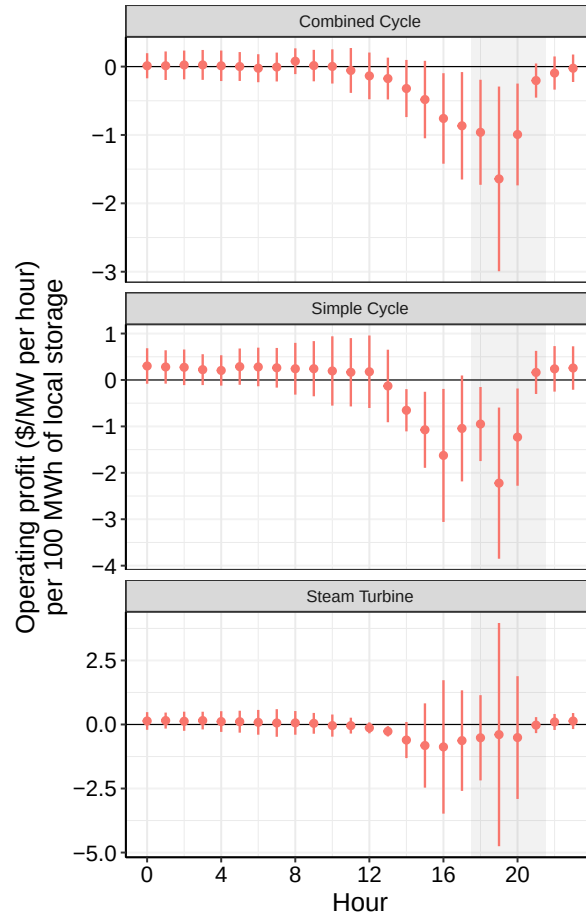
Notes. Binary event studies of the markup for midnight to 6 AM, 10 AM to 4 PM, and 5 PM to 9 PM, with years relative to the first local storage entry binned at three years and the year before entry as the reference. The 95% confidence intervals use t_{G-1} critical values and with standard errors clustered at the county level.

Operating profit and rents by technology Figure C.13 presents the hourly operating profit effects by technology. Combined cycle and simple cycle generators lose operating profit in the evening, by \$0.93 and \$1.06 per MW per hour on average from 5 PM to 9 PM. Simple cycle generators also gain a small amount overnight. The steam turbine estimates are imprecise. Table C.5 presents the daily effects by technology. Under the central benchmark, the market power rent falls by \$6.48 per MW-day for combined cycle and \$6.60 per MW-day for simple cycle generators, and the competitive rent does not change significantly. For steam turbines, the operating profit effect is not significant.

Market power rent event study Figure C.14 presents the event study for the daily market power rent. The coefficients before entry are small and insignificant. After entry, the market power rent falls by \$6 to \$11 per MW-day in the year of entry and the two years after it. The coefficient three or more years after entry is positive, and all post-entry coefficients are imprecise.

Alternative competitive benchmark Table C.6 reports the decomposition under a third benchmark, which sets the competitive price to the highest marginal cost among gas generators actually running. This benchmark attributes only 15 percent of the profit loss to market power, and the market power rent is not significant. It is, however, mechanically biased by the treatment. Storage discharge displaces exactly the most expensive running generators, so the benchmark falls when storage enters, and the decomposition records that fall as a loss of competitive rent. I therefore use the price-setter benchmark as the lower bound in the main text. Figure C.15 presents the hourly rents under the price-setter benchmark, which follow the same pattern as under the central benchmark.

Figure C.13: Effect of local storage on hourly operating profit by technology



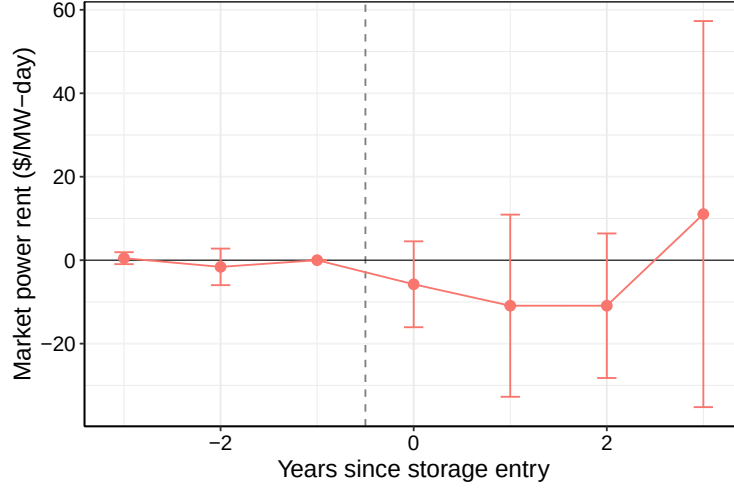
Notes. Two-stage difference-in-differences estimates of the effect of 100 MWh of local storage on hourly operating profit per MW of nameplate capacity, by technology, weighted by capacity. The 95% confidence intervals use t_{G-1} critical values and with standard errors clustered at the county level.

Table C.5: Effect of local storage on daily operating profit and rents by technology

	Operating profit	Competitive rent	Market power rent	Competitive rent (price-setter)	Market power rent (price-setter)
	(1)	(2)	(3)	(4)	(5)
<i>Combined Cycle</i>					
\$/MW-day per 100 MWh	-6.52** (2.69)	-0.05 (0.47)	-6.48** (2.51)	-1.95 (2.55)	-4.57* (2.35)
Share of profit effect	1.00	0.01	0.99	0.30	0.70
Annual total (\$ million)	-37.8 (15.6)	-0.3 (2.7)	-37.5 (14.5)	-11.3 (14.8)	-26.5 (13.6)
Per 100 MWh of storage (\$ thousand/year)	-3,193 (1,317)	-23 (231)	-3,170 (1,228)	-957 (1,251)	-2,236 (1,152)
<i>Simple Cycle</i>					
\$/MW-day per 100 MWh	-6.42** (2.65)	0.18 (1.51)	-6.60*** (2.10)	-0.79 (1.91)	-5.64** (2.68)
Share of profit effect	1.00	-0.03	1.03	0.12	0.88
Annual total (\$ million)	-13.6 (5.6)	0.4 (3.2)	-14.0 (4.5)	-1.7 (4.1)	-11.9 (5.7)
Per 100 MWh of storage (\$ thousand/year)	-1,151 (474)	32 (271)	-1,182 (376)	-141 (343)	-1,010 (480)
<i>Steam Turbine</i>					
\$/MW-day per 100 MWh	-3.49 (5.70)	-2.89*** (0.92)	-0.60 (5.44)	-1.57 (1.02)	-1.91 (4.87)
Share of profit effect	1.00	0.83	0.17	0.45	0.55
Annual total (\$ million)	-3.0 (4.9)	-2.5 (0.8)	-0.5 (4.6)	-1.3 (0.9)	-1.6 (4.2)
Per 100 MWh of storage (\$ thousand/year)	-252 (412)	-209 (67)	-43 (393)	-114 (74)	-138 (352)
Observations (generator-days)	206,025				
Counties	33				

Notes. Two-stage difference-in-differences estimates on the generator-day panel by technology, weighted by capacity. Columns are defined as in Table 6. Standard errors clustered by county in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure C.14: Event study: daily market power rent



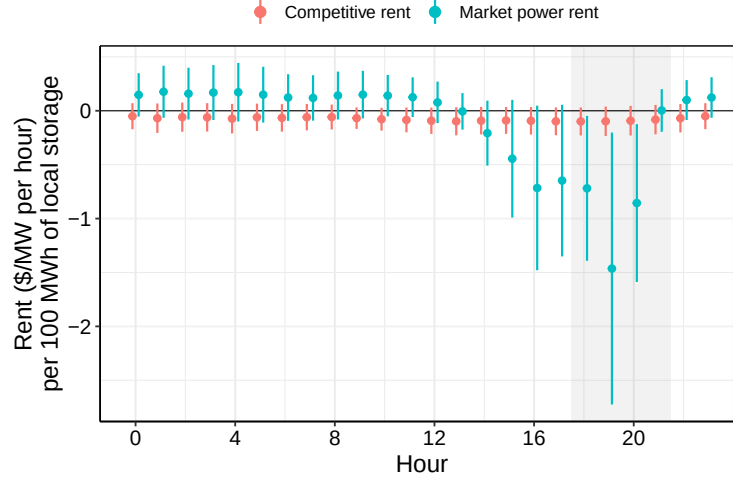
Notes. Binary event study of the daily market power rent per MW of capacity under the central benchmark, weighted by capacity, with the year before entry as the reference. The 95% confidence intervals use t_{G-1} critical values and with standard errors clustered at the county level.

Table C.6: Rent decomposition under the highest-running-cost benchmark

	Operating profit (1)	Competitive rent (highest running cost) (2)	Market power rent (highest running cost) (3)
\$/MW-day per 100 MWh	-5.33*** (1.86)	-4.51** (2.11)	-0.83 (2.57)
Share of profit effect	1.00	0.85	0.15
Annual total (\$ million)	-46.8 (16.3)	-39.5 (18.5)	-7.2 (22.5)
Per 100 MWh of storage (\$ thousand/year)	-3,950 (1,374)	-3,338 (1,563)	-612 (1,903)
Observations (generator-days)	677,117		
Counties	50		

Notes. Two-stage difference-in-differences estimates on the generator-day panel, weighted by capacity. The competitive price is the highest marginal cost among gas generators running in the local market and interval. Standard errors clustered by county in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

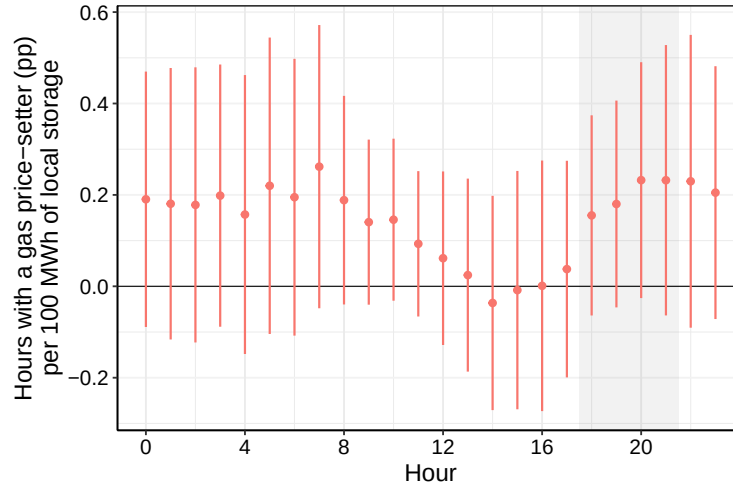
Figure C.15: Hourly rents under the price-setter benchmark



Notes. Two-stage difference-in-differences estimates of the effect of 100 MWh of local storage on the competitive and market power rents under the price-setter benchmark, per MW of capacity per hour, weighted by capacity. The 95% confidence intervals use t_{G-1} critical values and with standard errors clustered at the county level.

Gas price-setting Figure C.16 presents the effect of local storage on the probability that a gas generator sets the local price. Before storage entry, a gas generator sets the price in 53 to 69 percent of intervals, depending on the hour. The effect of storage is small and not significant in any hour, ranging from -0.04 to 0.23 percentage points per 100 MWh.

Figure C.16: Effect of local storage on gas price-setting

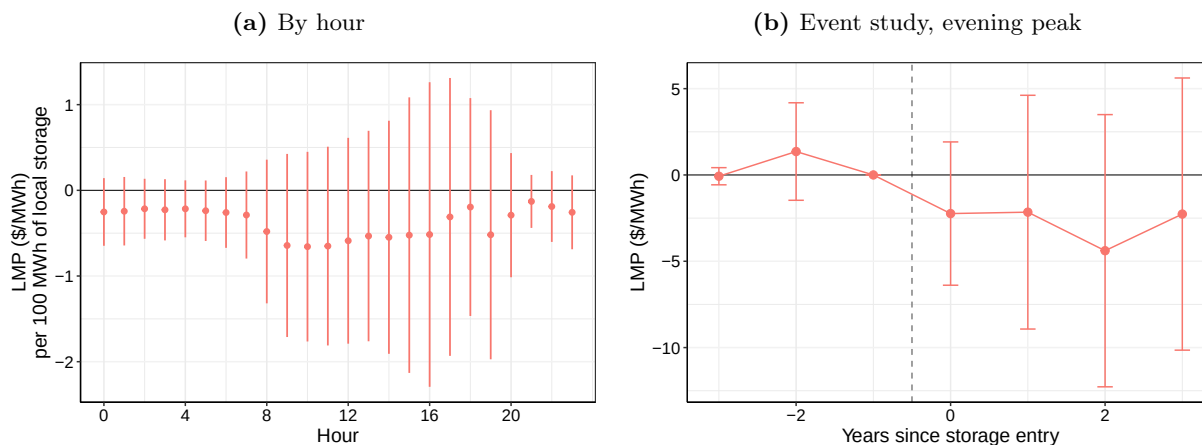


Notes. Two-stage difference-in-differences estimates of the effect of 100 MWh of local storage on the share of intervals in the hour in which a gas generator sets the local price, in percentage points. The 95% confidence intervals use t_{G-1} critical values and with standard errors clustered at the county level.

C.3 Additional Welfare Results

Effect on LMP Figure C.17 presents the effect of local storage on the locational marginal price at gas nodes, which enters the consumer surplus calculation in Equation (19). Panel (a) shows that the price falls in every hour, by \$0.13 to \$0.66 per MWh per 100 MWh of storage, with the largest declines in the late morning. None of the hourly effects is individually significant. Panel (b) presents the event study for the evening peak. The coefficients before entry are small and insignificant. After entry, the evening price falls by \$2 to \$4 per MWh, although the individual coefficients are imprecise. I therefore treat the consumer surplus estimate as imprecise.

Figure C.17: Effect of local storage on prices at gas nodes

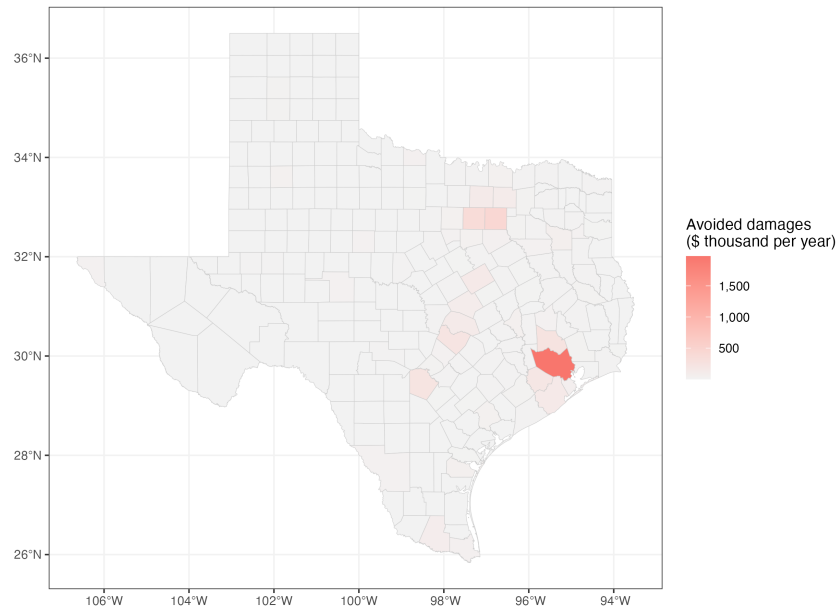


Notes. Two-stage difference-in-differences estimates on the node-by-hour panel of gas settlement points, in 2020 dollars per MWh, with settlement point and load-zone-by-date-by-hour fixed effects and the local renewable and temperature controls. Panel (a) reports the effect of 100 MWh of local storage by hour. Panel (b) reports a binary event study for 5 PM to 9 PM, with the year before entry as the reference. Intervals use t_{G-1} critical values and with standard errors clustered at the county level.

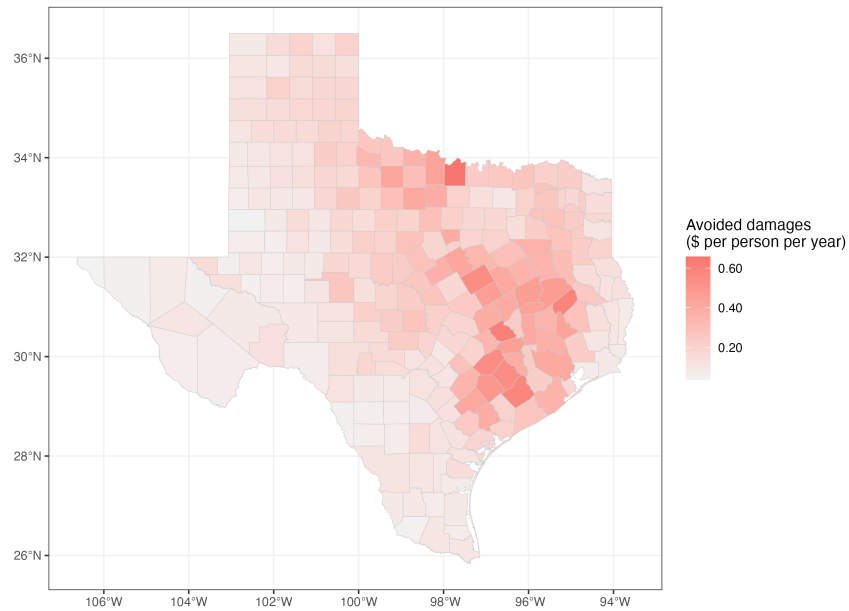
Distribution of avoided damages across Texas counties Figure C.18 maps the avoided local damages across Texas counties. Panel (a) shows the total avoided damages in each receptor county, which are largest in the populous counties of Harris, Dallas, Tarrant, and Bexar. Panel (b) divides the damages by county population. Per resident, the avoided damages are more evenly spread and remain highest in and around the counties that host gas generation and storage.

Figure C.18: Avoided local damages by Texas county

(a) Total



(b) Per capita



Notes. Annual avoided damages from NO_x , SO_2 , and $\text{PM}_{2.5}$ in each receptor county, in 2020 dollars, from the avoided emissions of gas generators displaced by local storage and the AP4 source-receptor matrices. Panel (b) divides by county population. The estimates use the local displacement and are an upper bound if displaced gas generation moves to counties without storage.

D Additional Tables

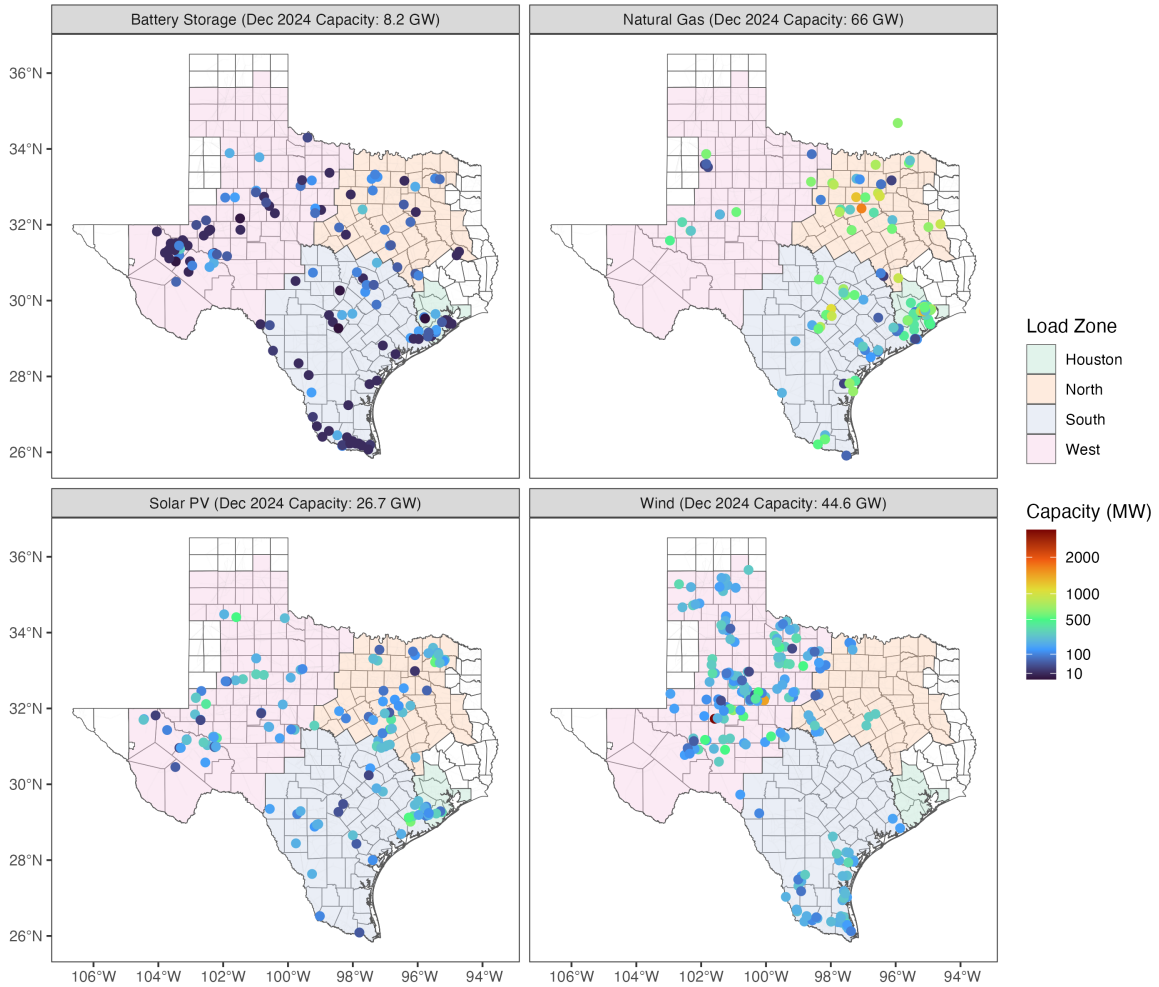
Table D.7: Generation Mix Summary

	Capacity			Mean Generation		Units	
	GW	Duration (hr)	%	GWh	%	Count	%
<i>Panel A: Fuel Type</i>							
Natural Gas	65.23		38.9	20.482	44.5	159	24.2
Coal	20.19		12.1	8.252	17.9	17	2.6
Wind	41.57		24.8	10.453	22.7	192	29.2
Solar PV	26.63		15.9	2.062	4.5	133	20.2
Batteries	8.04	1.33	4.8	0.043	0.1	142	21.6
Nuclear	5.23		3.1	4.666	10.1	2	0.3
Other	0.64		0.4	0.079	0.2	12	1.8
Total	167.54		100.0	46.036	100.0	657	100.0
<i>Panel B: Top 5 Natural Gas Firms</i>							
Luminant	12.91		19.8	4.767	21.7	18	11.3
Calpine Corporation	10.04		15.4	6.877	31.3	17	10.7
Constellation Energy	6.09		9.3	2.007	9.1	10	6.3
NRG Energy	5.64		8.6	0.943	4.3	12	7.6
City of San Antonio	5.53		8.5	1.449	6.6	10	6.3
Others	25.03		38.4	5.943	27.0	92	57.9
Total	65.23		100.0	21.985	100.0	159	100.0
<i>Panel C: Top 5 Storage Firms</i>							
Engie S.A.	1.83	1.02	22.8	0.017	27.8	24	16.9
Enel S.p.A.	1.19	1.69	14.8	0.014	22.6	15	10.6
Plus Power	0.79	1.92	9.8	0.009	14.8	4	2.8
Key Capture Energy	0.63	1.83	7.8	0.002	3.5	11	7.8
Luminant	0.44	2.05	5.5	0.003	5.4	4	2.8
Others	3.16	1.22	39.3	0.016	25.8	84	59.1
Total	8.04		100.0	0.061	100.0	142	100.0

Notes. Panel A show capacity, generation, and number of units by fuel type. Panel B show capacity, generation, and number of units by firms for natural gas unit, while Panel C show specifically for storage firms. Discrepancies in generation between each panel arise from differences in operating intervals when aggregating generation over multiple units. I define capacity using maximum observed generation from my data, this creates discrepancies in generation capacity compared to (ERCOT, 2025). The largest firm in ERCOT owns 15 percent of total capacity and contributes, on average, about 20 percent of electricity generation in each interval. The second to fifth largest firms own between 4 to 6 percent of total capacity, contributing on average between 5 to 14.6 percent of electricity generation. For storage firms, the largest owns 21.9 percent of storage capacity, while the second to fifth largest own between 3.6 to 10 percent of storage capacity. Based on just the top five storage firms, there is significant heterogeneity in storage characteristics in terms of unit capacity and duration.

E Additional Figures

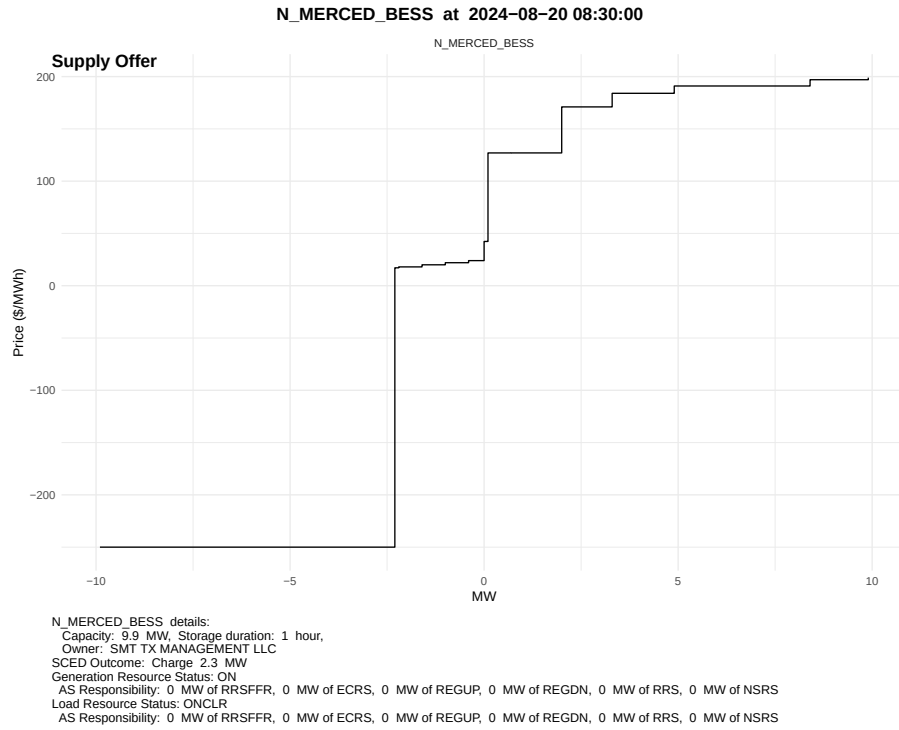
Figure E.19: Location of Generator in ERCOT



Notes: This figure shows the spatial distribution of each generator unit based on their fuel type in ERCOT. Natural gas and coal units are concentrated in the populated area of Texas, where the demand center is, such as the cities of Austin, Dallas, Houston, and San Antonio. Solar PV and storage are relatively spread evenly throughout the state, while wind is concentrated in the West and South load zone, the area with the highest wind potential.

Figure E.20: Example Storage Offer

(a) Example of unconstrained storage offer



(b) Example of energy constrained storage offer

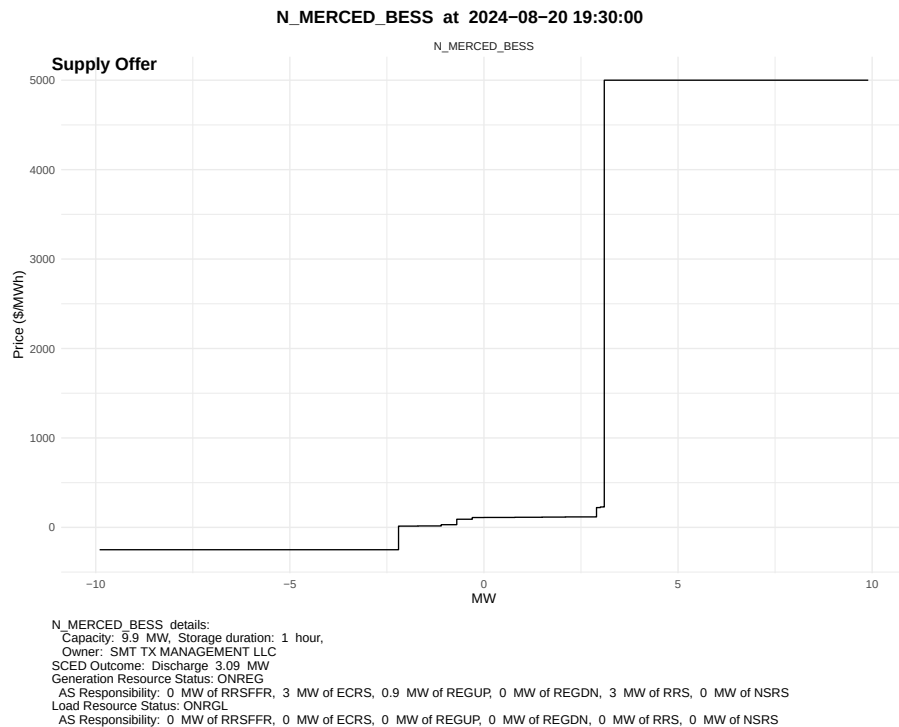


Figure E.21: Installed capacity by Fuel Type from 2018-2024

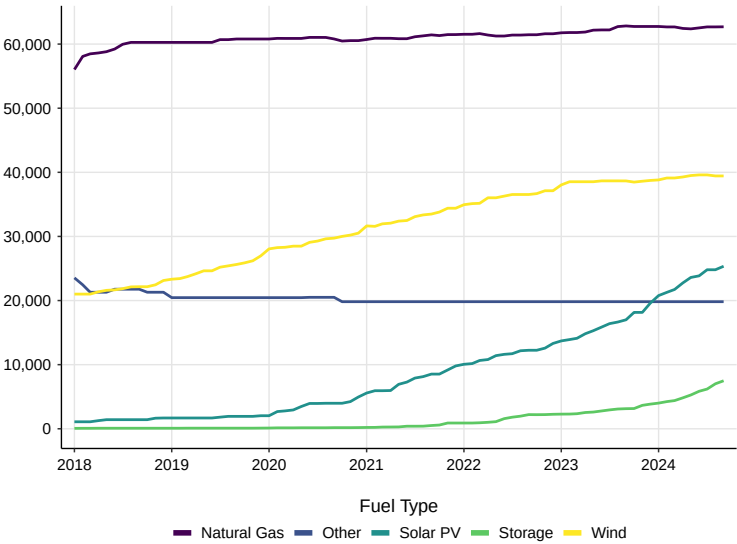
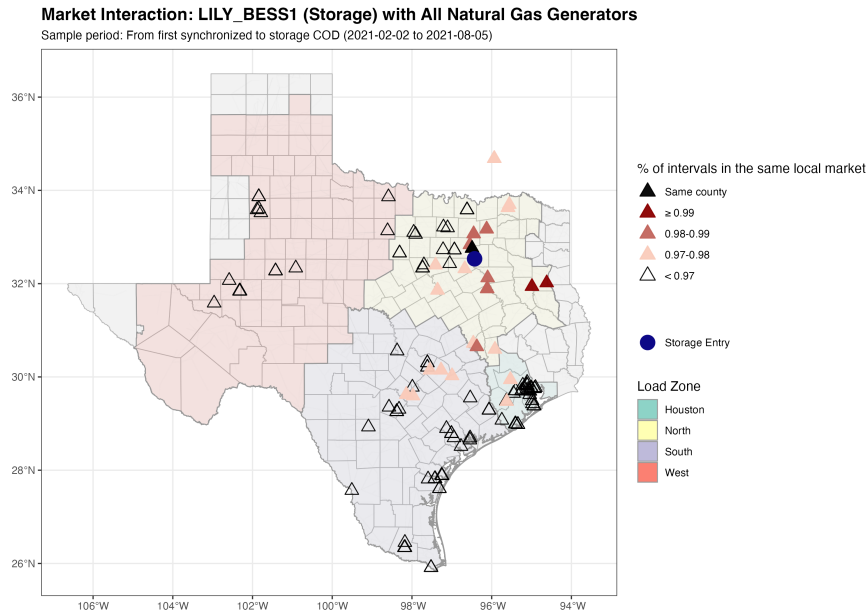
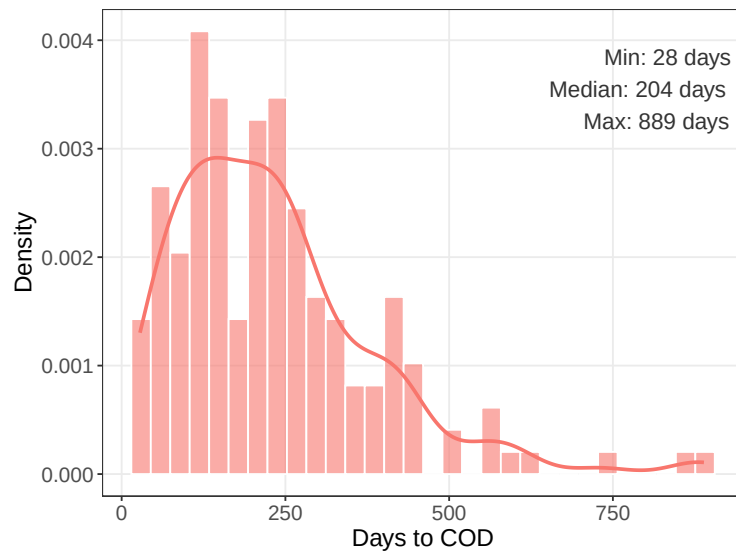


Figure E.22: Treatment assignment and market interaction frequency



Notes: This figure shows which natural gas generators are treated by the LILY_BESS1 storage unit. The blue dot shows the location of the storage unit. The solid black triangle represents generators located in the same county as the storage unit. The red triangle represents generators sharing the same local market with the storage unit more than 97% of the intervals prior to storage COD. The void triangle represents generators that do not share the same local market with the storage unit more than 97% of the intervals prior to storage COD.

Figure E.23: Distribution of storage first observation to COD



Notes: This figure shows the distribution of the time gap between the first date a storage unit shows up in the SCED data (synchronized) and the COD date.